A Biobjective Model-Driven Autocoder for Blind Hyperspectral Unmixing

Jianjun Liu, *Member, IEEE*, Hongru Zong, Zebin Wu, *Senior Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*— Hyperspectral unmixing (HU) decomposes hyperspectral images (HSIs) into pure spectral signatures (endmembers) and their proportions (abundances). Existing deep learning methods for this task predominantly focus on either linear or nonlinear relationships, often struggling to achieve an optimal equilibrium between the robustness of linear estimations and the precision of nonlinear models. Furthermore, these methods often neglect the integration of spatial information and the interpretability of the results. This article proposes a biobjective model-driven autoencoder network for blind HU that simultaneously addresses both linear and nonlinear relationships. By combining linear and nonlinear kernel models within a model-driven deep learning framework, we aim to enhance the interpretability of the results. To effectively capture spatial information, we introduce an adaptive composite kernel (ACK) that integrates traditional and spatial–spectral kernels (SSKs), using a dynamic convolution-like mechanism to optimize their respective weights. Our method establishes linear and nonlinear reconstruction losses, enabling the simultaneous estimation of endmembers from both types of relationships, thus capturing the complex and high-dimensional data structures inherent in HSIs. Experiments on real and synthetic datasets demonstrate the superiority of the proposed method.**

***Index Terms*— Autocoder network, blind hyperspectral unmixing (HU), hyperspectral images (HSIs), kernel-based methods.**

I. Introduction

HYPERSPECTRAL images (HSIs) are cube-like structures that capture both rich spatial details and a vast range of narrow spectral bands, offering a more comprehensive representation of the scene compared to RGB images. Due to their rich information content, HSIs find wide applications in many fields [1], [2], [3], [4], [5]. However, due to the limited spatial resolution of sensors and the potential mixing of different materials, HSIs often contain mixed pixels. To address this issue, the task of hyperspectral unmixing (HU) has gained increasing attention. HU aims to decompose HSIs into pure spectral signals (endmembers) and their corresponding proportions (abundances) [6]. Based on the causes of pixel mixing, existing unmixing methods are generally categorized into linear unmixing and nonlinear unmixing approaches [7], [8], [9], [10].

The linear mixing model (LMM) is extensively utilized in the field of HU due to its inherent simplicity and computational efficiency [6]. This model posits that the spectral signature observed at each pixel within an HSI can be represented as a linear combination of the spectral signatures of the constituent pure materials, with the weights corresponding to their respective abundance fractions. Traditional methods typically involve two steps: endmember extraction and abundance estimation. First, under the assumption of pure pixels, endmembers are extracted using geometric methods such as pixel purity index [11], N-FINDR [12], vertex component analysis (VCA) [13], and the simplex growing algorithm [14]. Subsequently, techniques like fully constrained least squares [15] and sparse unmixing by variable splitting and augmented Lagrangian [16] are employed for abundance estimation. However, since the fundamental assumption of pure pixels is often not met in most images, independently performing these steps leads to suboptimal results. As a result, there is a growing trend toward blind unmixing methods that simultaneously estimate both endmembers and abundances. Some traditional statistical algorithms are widely applied in this regard, including nonnegative matrix factorization (NMF) [17], [18], [19], [20], [21] and Bayesian methods [22], [23].

However, the LMM is not adequately suited for HSIs of complex Earth surfaces, as light undergoes multiple interactions with pure substances prior to reaching the sensor, leading to a nonlinear relationship between pixel signatures and endmember spectra. As a result, nonlinear unmixing techniques have been developed to more effectively tackle these challenges [24], [25]. For example, the Hapke model [26] is widely used in intimate mixture models and effectively describes the inherent nonlinear relationships between individual endmembers, particularly in dense materials, by considering factors such as surface roughness and light interaction under complex environmental conditions. However, obtaining accurate parameters can be challenging, making it difficult to ensure precision. Among the various nonlinear mixing methods employed for HU, bilinear mixing models are the most prevalent. In these models, multiple scattering effects, primarily arising from second-order scattering, are often simplified by

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Jianjun Liu and Hongru Zong are with Jiangsu Provincial Engineering Laboratory for Pattern Recognition and Computational Intelligence, School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi 214122, China (e-mail: liuofficial@163.com; zonghr123@163.com).

Zebin Wu and Liang Xiao are with the School of Computer Science, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: zebin.wu@gmail.com; xiaoliang@mail.njust.edu.cn).

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disregarding higher order scattering components and other scattering factors. The Fan model [27], which is empirically derived from experimental data, accurately represents the interactions occurring between two endmembers, exhibiting a correlation with their respective abundances. The generalized bilinear model (GBM) [28] bridges the gap between the Fan model and LMM. The posterior polynomial mixing model [29] further extends the bilinear model by incorporating posterior polynomials, building upon the GBM model to enhance interpretability. However, bilinear models merely account for second-order scattering effects. To address this limitation, Heylen and Scheunders [30] introduced multilinear mixing models (MLMMs), which take into consideration the effects of multiple scattering interactions among different materials. Nonetheless, the implementation of this model results in increased computational complexity, and the estimation of interaction probabilities presents further challenges. Notably, kernel-based methods have gained widespread application in the field of nonlinear unmixing due to their ability to map nonlinear spaces into linear counterparts. Chen et al. [31] defined function spaces within the framework of reproducing kernel Hilbert spaces and selected appropriate kernel functions to accurately capture the nonlinear relationships between endmembers. Additionally, Huang et al. [18] demonstrated the effectiveness of incremental kernel NMF for HU, where kernel-based methods were employed to model complex nonlinearities, thereby improving unmixing accuracy and computational efficiency. Similarly, Gu et al. [32] proposed a kernel-transformed bilinear mixing model for nonlinear HU, which reformulates traditional bilinear models into their kernel-based equivalents, further emphasizing the ability of kernel methods to capture nonlinear interactions in high-dimensional hyperspectral data.

In recent years, deep learning methods have shown remarkable performance in the field of computer vision, including significant progress in hyperspectral tasks [33], [34], [35], [36], [37]. At the same time, there has been a growing trend in utilizing autoencoder-based deep learning methods for HU tasks. Autoencoders are especially well-suited for HU due to their strengths in feature extraction, dimensionality reduction, and unsupervised learning [38]. For instance, Qu and Qi [39] proposed a sparse unit denoising autoencoder (uDAS) to improve robustness. Meanwhile, applications of convolutional autoencoders are also on the rise. Palsson et al. [40] processed images into patches, utilizing convolution to extract spatial features while simultaneously estimating abundances and endmembers. The cycle-consistency unmixing network (CyCU-Net) [41] employs two cascaded autoencoders to enhance unmixing capabilities. Rasti et al. [42] introduced minimum volume as a regularization term to aid the network in learning geometric features. Xu et al. [43] proposed a convolutional autoencoder based on temperature scaling to apply different sparse constraints to different regions of the abundance. Tao et al. [44] proposed a dual-feature fusion network that enhances HU by calculating spectral and spatial similarities through a feature fusion module, while decoupling endmember and abundance estimation into dedicated submodules for joint optimization. Unlike traditional convolutional methods, the self-attention mechanism [45] excels at capturing global dependencies, thus enabling a more effective extraction of holistic image features. The self-attention methods have gained popularity in HU. Yang et al. [46] improved the extraction of both global and detailed information by utilizing a U-shaped Swin-Transformer network. Another study [47] introduced a double-aware Transformer model, which enhances the ability to extract region homogeneity and spectral correlation information. Tao et al. [48] proposed a abundance-guided attention spectral and spatial attention network (A2SAN) to directly extract endmembers and abundances from images without the need for endmember initialization. Wang et al. [49] proposed a multiscale aggregation Transformer network that captures features at multiple scales and integrates a Transformer encoder for efficient feature fusion. The integration of traditional optimization methods with deep learning has garnered significant attention. For instance, Li et al. [50] proposed a self-supervised robust deep NMF model that overcomes the drawbacks of conventional NMF through an asymmetric encoder–decoder architecture, effectively reducing noise. Qian et al. [51] introduced an approach that integrates the iterative shrinkage-thresholding algorithm with deep learning techniques to solve the blind unmixing challenge. Xiong et al. [52] introduced a method derived from the unrolling of ℓ_p-constrained NMF optimization. Zhou and Rodrigues [53] proposed an encoder to solve linear objective problems using the alternating direction method of multipliers (ADMM), thereby enhancing the network's physical interpretability. However, these methods often struggle to effectively integrate the stability offered by traditional optimization techniques with the robust feature extraction capabilities of deep neural networks, especially when they come to extracting intricate spatial patterns and nonlinear data relationships. Recent research has demonstrated the growing relevance of diffusion models in a wide range of HSI applications [54]. Likewise, these models have also found utility in the field of HU. Deng et al. [55] proposed a diffusion model-based semi-blind unmixing method, which leverages spectral prior knowledge from large libraries. However, despite these advancements, deep learning models that rely on linear mixing assumptions encounter significant limitations. Linear models frequently fail to adequately capture the intricate nonlinear relationships and interactions present within hyperspectral data. This limitation highlights the necessity for more advanced deep learning techniques that are capable of addressing the complexities inherent in HU.

Considering the aforementioned issues, deep learning methods strive to capture the complex nonlinear relationships among endmembers with enhanced precision. For instance, some generic mixing models integrate both linear and additive nonlinear mixing components. Li et al. [56] regarded mixing models as nonlinear fluctuations superimposed on linear mixing, thereby imposing specific structures on the encoder and decoder networks. Likewise, the HapkeCNN [57], based on

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the quadratic terms of the Hapke model, aimed to capture nonlinear features of images but encountered challenges in parameter acquisition. Additionally, some studies [58], [59] decomposed HU into linear and bilinear quadratic scattering parts, proposing simulated hierarchical bilinear mixing models. Shahid and Schizas [60] designed a decoder that incorporates three bilinear models to boost model versatility. Nevertheless, these models have not fully leveraged the abundant spatial information in HSIs. Meanwhile, MLMM has gained widespread application in deep learning for HU. Fang et al. [61] developed a decoder that aligns with multilinear models to simulate multilinear mixing, implementing blind nonlinear unmixing under MLMM using an enhanced deep autoencoder network (MLM-3DAE). However, deep learning methods based on MLMM still face challenges, such as the increased complexity in decoder design and the growth in parameter count, which may reduce unmixing efficiency.

Notably, kernel-based unmixing methods require fewer parameters to effectively map data to high-dimensional spaces, making them more suitable for modeling nonlinear relationships among spectral components. By leveraging these advantageous properties, the integration of kernel-based unmixing methods with deep learning is anticipated to effectively capture the complex relationships between endmembers in HSIs. Recent studies have explored such integration, including the work by Xing et al. [62], who introduced a deep encoder with kernel-wise Taylor series for HSI classification, and Xi et al. [63], who proposed a multiscale context-aware ensemble deep kernel extreme learning machine for efficient HSI classification. These methods highlight the benefits of combining kernel-based techniques with deep learning to enhance feature extraction and classification accuracy. Although this strategy holds potential advantages, it still faces challenges in three aspects.

1) Existing methods have not yet explored how to integrate kernel-based unmixing methods with deep learning for HU, and like general nonlinear deep unmixing methods, the issue of how to incorporate spatial information remains unresolved.
2) Current nonlinear unmixing methods do not consider the interpretability of networks, which is crucial for understanding and trusting the results.
3) Existing deep learning methods typically employ either linear or nonlinear models alone, leading to difficulties in achieving a balance between robustness and precision.

To address these issues, we propose a novel biobjective autoencoder-based deep learning method for blind unmixing of HSIs. Unlike conventional approaches that focus solely on either linear or nonlinear unmixing, our method simultaneously models mixing effects within a unified framework, ensuring a more comprehensive and accurate representation of HSIs. One objective employs a linear model to guarantee fundamental solutions, while the other objective leverages a nonlinear kernel model to capture complex and high-dimensional data structures inherent in HSIs. Our network comprises a physically interpretable encoder, guided by the biobjective model, and a decoder that is tailored to tackle both linear unmixing and nonlinear unmixing simultaneously. In contrast to traditional deep learning methods that rely on end-to-end training to optimize parameters, our network explicitly integrates the ADMM framework, updating parameters through a series of well-defined optimization steps. This integration not only makes the parameter update process more transparent, providing a sharp contrast to the typically closed-box nature of conventional deep learning methods but also offers greater interpretability in understanding how parameters are adjusted during optimization. To enhance the integration of spatial information, we introduce an adaptive composite kernel (ACK) within the kernel computation process. This ACK seamlessly integrates spatial–spectral kernel (SSK) [64] and traditional kernel and dynamically adjusts their respective weights during the computation. By optimizing the balance between these kernels, our method is able to more effectively and efficiently extract spatial features from HSIs, thereby enhancing the overall performance of the blind unmixing process. The specific contributions are as follows.

1) We propose a biobjective model-driven autoencoder network, designed to concurrently tackle both linear and nonlinear challenges inherent in HSIs. The contributions of the linear and nonlinear components are dynamically balanced through the weighted adjustment.
2) We devise the encoder structure by unrolling the ADMM algorithm and concurrently develop the decoder to handle linear and nonlinear problems, enhancing the model's interpretability. The decoder independently learns the endmembers for each objective and updates them synchronously via the loss function.
3) We employ an ACK that integrates SSK and traditional kernel to enhance spatial information extraction. Utilizing a dynamic convolution-like mechanism, we adaptively adjust the weights assigned to each kernel, thereby ensuring an optimal and equilibrated distribution of their contributions.

II. Biobjective Unmixing Model

Most unmixing methods primarily address single-objective optimization problems. However, the exclusive reliance on a single objective frequently inadequately captures the complexities of HU, highlighting the necessity for a more holistic approach [65]. In this section, we introduce a biobjective model tailored for HU and subsequently solve it using the ADMM. The biobjective optimization problem comprises two fundamental components. One component involves a linear optimization process within the original data space. The other component entails an optimization task that is projected onto an extended, high-dimensional feature space by utilizing a kernel function.

The LMM posits that mixed pixels result from linear combinations of endmembers. Within the LMM framework, HU can be mathematically formulated as follows:

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{\Gamma} \tag{1}$$

where $\mathbf{Y} = (\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N) \in \mathbb{R}^{L\times N}$ is the observed HSI with L-bands and N pixels, and $\mathbf{E} = (\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_P) \in \mathbb{R}^{L\times P}$

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represents an endmember matrix containing P endmembers. The matrix $\mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_N) \in \mathbb{R}^{P\times N}$ serves as the abundance matrix, and $\mathbf{\Gamma} \in \mathbb{R}^{L\times N}$ represents the noise matrix. Endmembers adhere to the nonnegativity constraint, as their respective curves represent the reflectivity characteristics of various materials [66], i.e., $\mathbf{E} \geq 0$. Simultaneously, abundance denotes the fraction or contribution of each endmember present within a mixed pixel. The abundance matrix is subject to two key constraints: the abundance nonnegativity constraint (ANC), i.e., $\mathbf{A} \geq 0$, and the abundance sum-to-one constraint (ASC), i.e., $\mathbf{1}_P^T\mathbf{A} = \mathbf{1}_N^T$. Here, $\mathbf{1}_P$ denotes a column vector composed of P elements with all being 1. These two constraints guarantee that the abundance values possess physical significance, ensuring that they are nonnegative and their summation equals one.

In the proposed biobjective model, the linear objective function to be minimized is expressed as follows:

$$J_\chi(\mathbf{E}, \mathbf{A}) = \frac{1}{2}\|\mathbf{Y} - \mathbf{EA}\|_\mathbf{F}^\mathbf{2} \tag{2}$$

where $\|\cdot\|_F^2$ represents the Frobenius norm and χ represents the original input space. The other objective function to be minimized in the mapped feature space is as follows:

$$J_\mathcal{H}(\mathbf{E}, \mathbf{A}) = \frac{1}{2}\|\Phi(\mathbf{Y}) - \Phi(\mathbf{E})\mathbf{A}\|_F^2 \tag{3}$$

where $\Phi(\cdot)$ is a feature mapping function for matrices and $\mathcal{H}$ represents the feature space after mapping by the kernel function. In general, it is not possible to achieve optimal solutions for both objectives simultaneously. Therefore, we introduce a balancing factor $\alpha > 0$ to trade off between them, aiming to achieve a Pareto optimal solution

$$\min \alpha J_\chi(\mathbf{E}, \mathbf{A}) + (1-\alpha)J_\mathcal{H}(\mathbf{E}, \mathbf{A}). \tag{4}$$

For convenience, we reformulate the target problem at the pixel level and introduce an ℓ_1 norm sparsity term to regulate the abundance, thereby expanding the biobjective problem into the following form:

$$\begin{aligned}&\min_{\mathbf{E},\mathbf{a}} \frac{\alpha}{2}\|\mathbf{y} - \mathbf{Ea}\|_2^2 + \frac{1-\alpha}{2}\|\phi(\mathbf{y}) - \Phi(\mathbf{E})\mathbf{a}\|_2^2 + \lambda\|\mathbf{a}\|_1\\ &\text{s.t. } \mathbf{E} \geq 0, \quad \mathbf{a} \geq \mathbf{0}, \ \mathbf{1}_P^T\mathbf{a} = 1\end{aligned} \tag{5}$$

where $\mathbf{y} \in \mathbb{R}^L$, $\mathbf{a} \in \mathbb{R}^P$, $\|\cdot\|_q$ $(q = 1, 2)$ denotes ℓ_q norm, $\phi(\cdot)$ is a feature mapping function for vectors, and $\lambda > 0$ is a balance parameter.

To address the optimization problem (5), an alternating update approach can be employed to solve for $\mathbf{a}$ and $\mathbf{E}$. The solving of $\mathbf{a}$ is a complex optimization problem, while the solving of $\mathbf{E}$ is a reconstruction problem. In this article, we consider solving them separately during the encoding and decoding phases, specifically solving for $\mathbf{a}$ in the encoding phase and for $\mathbf{E}$ in the decoding phase. Moreover, to facilitate effective exploration of the solution space of $\mathbf{a}$, the ASC constraint is dropped during its optimization process but kept in the final stage of decoder. Next, we focus solely on solving for the problem $\mathbf{a}$, leaving others to Section III.

Without the ASC constraint, the $\mathbf{a}$-subproblem of (5) can be written as follows:

$$\min_\mathbf{a} \frac{\alpha}{2}\|\mathbf{y} - \mathbf{Ea}\|_2^2 + \frac{1-\alpha}{2}\|\phi(\mathbf{y}) - \Phi(\mathbf{E})\mathbf{a}\|_2^2 + f(\mathbf{a}) \tag{6}$$

where $f(\mathbf{a}) = \lambda\|\mathbf{a}\|_1 + \iota_{\mathbb{R}^+}(\mathbf{a})$ represents a regularization term. $\iota_{\mathbb{R}^+}(\mathbf{a})$ represents the indicator function, defined as follows:

$$\iota_{\mathbb{R}^+}(\mathbf{a}) = \begin{cases} 0, & \mathbf{a} \geq 0 \\ \infty, & \mathbf{a} < 0. \end{cases} \tag{7}$$

ADMM possesses the capability to decompose intricate problems into more tractable subproblems, thereby effectively managing complex constraints and mitigating computational expenses. Thus, we employ ADMM to address the problem (6). To this end, we introduce auxiliary variables $\mathbf{v}_1 = \mathbf{Ea} \in \mathbb{R}^L$, $\mathbf{v}_2 = \mathbf{a} \in \mathbb{R}^P$, and $\mathbf{v}_3 = \mathbf{a} \in \mathbb{R}^P$ and simultaneously introduce dual variables $\mathbf{d}_1 \in \mathbb{R}^L$, $\mathbf{d}_2 \in \mathbb{R}^P$, and $\mathbf{d}_3 \in \mathbb{R}^P$. The augmented Lagrange equation of (6) can be written as follows:

$$\begin{aligned}\min_{\mathbf{a},\mathbf{v}_1,\mathbf{v}_2,\mathbf{v}_3} &\frac{\alpha}{2}\|\mathbf{y} - \mathbf{v}_1\|_2^2 + \frac{1-\alpha}{2}\|\phi(\mathbf{y}) - \Phi(\mathbf{E})\mathbf{v}_2\|_2^2 + f(\mathbf{v}_3)\\ &+ \frac{\mu}{2}\|\mathbf{Ea} - \mathbf{v}_1 - \mathbf{d}_1\|_2^2 + \frac{\mu}{2}\|\mathbf{a} - \mathbf{v}_2 - \mathbf{d}_2\|_2^2\\ &+ \frac{\mu}{2}\|\mathbf{a} - \mathbf{v}_3 - \mathbf{d}_3\|_2^2\end{aligned} \tag{8}$$

where $\mu > 0$ is a penalty parameter. Then, (8) can be decomposed into four subproblems at the $(k+1)$th iteration as follows:

$$\begin{aligned}\mathbf{a}^{k+1} &= \arg\min_\mathbf{a} \left\|\mathbf{Ea} - \mathbf{v}_1^k - \mathbf{d}_1^k\right\|_2^2 + \left\|\mathbf{a} - \mathbf{v}_2^k - \mathbf{d}_2^k\right\|_2^2\\ &\quad + \left\|\mathbf{a} - \mathbf{v}_3^k - \mathbf{d}_3^k\right\|_2^2\\ \mathbf{v}_1^{k+1} &= \arg\min_{\mathbf{v}_1} \frac{\alpha}{2}\|\mathbf{y} - \mathbf{v}_1\|_2^2 + \frac{\mu}{2}\left\|\mathbf{Ea}^{k+1} - \mathbf{v}_1 - \mathbf{d}_1^k\right\|_2^2\\ \mathbf{v}_2^{k+1} &= \arg\min_{\mathbf{v}_2} \frac{1-\alpha}{2}\|\phi(\mathbf{y}) - \Phi(\mathbf{E})\mathbf{v}_2\|_2^2\\ &\quad + \frac{\mu}{2}\left\|\mathbf{a}^{k+1} - \mathbf{v}_2 - \mathbf{d}_2^k\right\|_2^2\\ \mathbf{v}_3^{k+1} &= \arg\min_{\mathbf{v}_3} f(\mathbf{v}_3) + \frac{\mu}{2}\left\|\mathbf{a}^{k+1} - \mathbf{v}_3 - \mathbf{d}_3^k\right\|_2^2.\end{aligned} \tag{9}$$

Each variable at the $(k+1)$th iteration can be solved as follows:

$$\begin{aligned}\mathbf{a}^{k+1} &= (\mathbf{E}^T\mathbf{E} + 2\mathbf{I})^{-1}\left[\mathbf{E}^T\left(\mathbf{v}_1^k + \mathbf{d}_1^k\right) + \left(\mathbf{v}_2^k + \mathbf{d}_2^k\right)\right.\\ &\quad \left.+\left(\mathbf{v}_3^k + \mathbf{d}_3^k\right)\right] &(10)\\ \mathbf{v}_1^{k+1} &= (\alpha+\mu)^{-1}\left(\alpha\mathbf{y} + \mu\mathbf{Ea}^{k+1} - \mu\mathbf{d}_1^k\right) &(11)\\ \mathbf{v}_2^{k+1} &= [(1-\alpha)K(\mathbf{E}, \mathbf{E}) + \mu\mathbf{I}]^{-1}\\ &\quad \times \left[(1-\alpha)K(\mathbf{E}, \mathbf{y}) + \mu\left(\mathbf{a}^{k+1} - \mathbf{d}_2^k\right)\right] &(12)\\ \mathbf{v}_3^{k+1} &= \max\left(\mathbf{a}^{k+1} - \mathbf{d}_3^k - \lambda/\mu, 0\right) &(13)\\ \mathbf{d}_1^{k+1} &= \mathbf{d}_1^k - \left(\mathbf{Ea}^{k+1} - \mathbf{v}_1^{k+1}\right) &(14)\\ \mathbf{d}_2^{k+1} &= \mathbf{d}_2^k - \left(\mathbf{a}^{k+1} - \mathbf{v}_2^{k+1}\right) &(15)\\ \mathbf{d}_3^{k+1} &= \mathbf{d}_3^k - \left(\mathbf{a}^{k+1} - \mathbf{v}_3^{k+1}\right) &(16)\end{aligned}$$

where $\mathbf{I}$ is an identity matrix, $K(\mathbf{E}, \mathbf{E}) = \Phi^T(\mathbf{E})\Phi(\mathbf{E}) \in \mathbb{R}^{P\times P}$ is a Gram matrix, and $K(\mathbf{E}, \mathbf{y}) = \Phi^T(\mathbf{E})\phi(\mathbf{y}) \in \mathbb{R}^P$. Here, $K(\mathbf{x}_i, \mathbf{x}_j) = \langle\phi(\mathbf{x}_i), \phi(\mathbf{x}_j)\rangle$ with $\langle\cdot,\cdot\rangle$ being the inner product. The Gaussian kernel, given by $K(\mathbf{x}_i, \mathbf{x}_j) =$

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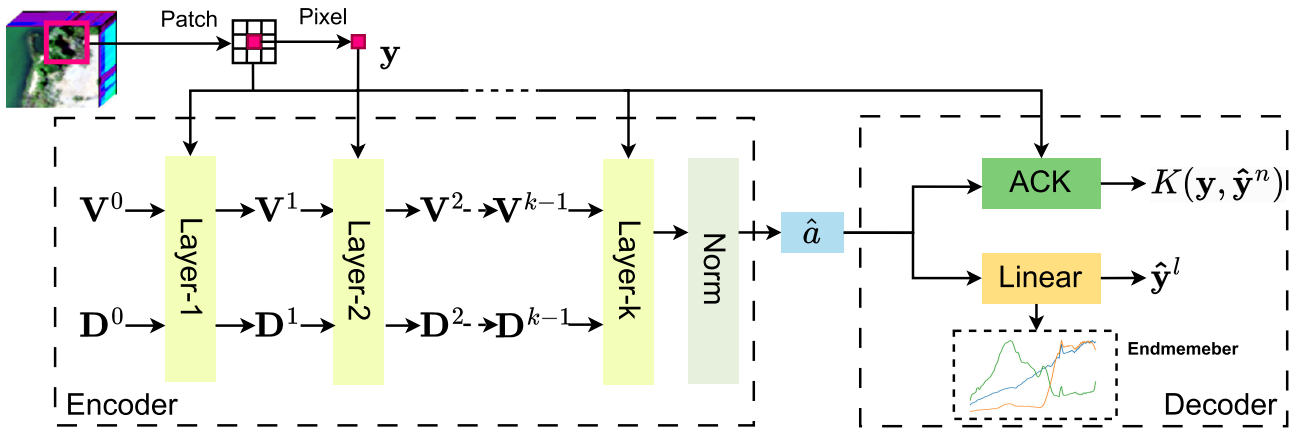


Fig. 1. Detailed structure of the proposed network, where $\mathbf{V}^k = \{\mathbf{v}_1^k, \mathbf{v}_2^k, \mathbf{v}_3^k\}$ and $\mathbf{D}^k = \{\mathbf{d}_1^k, \mathbf{d}_2^k, \mathbf{d}_3^k\}$.

$\exp(-(1/2\sigma^2)\|\mathbf{x}_i - \mathbf{x}_j\|^2)$, with σ controlling the width of the kernel, is used in the proposed model. The function $\max(\cdot, 0)$ in (13) is used to enforce the nonnegativity constraint by setting any negative values to zero.

III. Biobjective Model-Driven Autoencoder

Fig. 1 depicts the proposed network architecture. It is an interpretable and flexible network within the encoder–decoder framework. The encoder design is derived from the iterative process addressing the abundance problem **a** of (5). This design involves sequential updates of variables across each layer of the encoder. The decoder, on the other hand, is formulated based on the endmember problem **E** of (5). It comprises two distinct branches: a linear branch and a nonlinear branch. Unlike conventional approaches that separately handle linear and nonlinear mixing, our model seamlessly integrates both within a unified framework, allowing for a more accurate and comprehensive unmixing process. In particular, the nonlinear branch incorporates an ACK module to enhance its functionality and performance. The subsequent subsections will provide a detailed exploration of the network's design.

A. Encoder

The encoder is constructed through the deep unfolding of (10)–(16), with each layer corresponding to one iteration. Each layer is divided into three components: abundance update, auxiliary variable update, and dual variable update, as illustrated in Fig. 2.

The abundance update, responsible for updating the abundance in accordance with (10), operates in the $(k+1)$th layer of the neural network as follows:

$$\mathbf{a}^{k+1} = \mathbf{W}_1\left(\mathbf{v}_1^k + \mathbf{d}_1^k\right) + \mathbf{W}_2\left(\mathbf{v}_2^k + \mathbf{d}_2^k + \mathbf{v}_3^k + \mathbf{d}_3^k\right) \tag{17}$$

where $\mathbf{W}_1 \in \mathbb{R}^{P\times L}$ and $\mathbf{W}_2 \in \mathbb{R}^{P\times P}$ are learnable variables achieved by 1×1 convolutions. Here, $\mathbf{W}_1$ is initialized as $(\mathbf{E}^T\mathbf{E} + 2\mathbf{I})^{-1}\mathbf{E}^T$, and $\mathbf{W}_2$ is initialized as $(\mathbf{E}^T\mathbf{E} + 2\mathbf{I})^{-1}$. The trained weights evolve differently in each layer. The divergence allows the model to more accurately capture and represent the inherent characteristics of the underlying data [53].

The auxiliary variable update includes the updates of three auxiliary variables. According to (11), the update formula for $\mathbf{v}_1$ in the $(k+1)$th layer of the neural network is as follows:

$$\mathbf{v}_1^{k+1} = \theta_1\mathbf{y} + \mathbf{W}_3\mathbf{a}^{k+1} - \theta_2\mathbf{d}_1^k \tag{18}$$

where θ_1 and θ_2 are learnable parameters, initialized as $\alpha/(\alpha+\mu)$ and $\mu/(\alpha+\mu)$, respectively. $\mathbf{W}_3 \in \mathbb{R}^{L\times P}$ represents learnable variables, implemented as 1×1 convolutions and initialized as $\mu\mathbf{E}/(\alpha+\mu)$.

According to (12), the update formula for $\mathbf{v}_2$ in the $(k+1)$th layer of the neural network is as follows:

$$\mathbf{v}_2^{k+1} = \mathbf{W}_4 K(\mathbf{\Theta}, \mathbf{y}) + \mathbf{W}_5\left(\mathbf{a}^{k+1} - \mathbf{d}_2^k\right) \tag{19}$$

where $\mathbf{W}_4 \in \mathbb{R}^{P\times P}$ and $\mathbf{W}_5 \in \mathbb{R}^{P\times P}$ are learnable variables, achieved by 1×1 convolutions, and $\mathbf{\Theta} \in \mathbb{R}^{L\times P}$ is another learnable variable. The initialization of $\mathbf{W}_4$ and $\mathbf{W}_5$ is set to $(1-\alpha)[(1-\alpha)K(\mathbf{E},\mathbf{E})+\mu\mathbf{I}]^{-1}$ and $\mu[(1-\alpha)K(\mathbf{E},\mathbf{E})+\mu\mathbf{I}]^{-1}$, respectively, while $\mathbf{\Theta}$ is initialized as **E**. For the kernel[1] $K(\mathbf{\Theta}, \mathbf{y})$, it employs the ACK module, which combines the SSK and traditional kernel to effectively integrate spatial information, as detailed in Section III-B.

According to (13), the update formula for $\mathbf{v}_3$ in the $(k+1)$th layer of the neural network is as follows:

$$\mathbf{v}_3^{k+1} = \mathrm{ReLU}\left(\mathbf{a}^{k+1} - \mathbf{d}_3^k - \theta_3\right) \tag{20}$$

where θ_3 is a learnable parameter, initialized as λ/μ. The $\mathrm{ReLU}(\cdot)$, commonly used in neural networks, not only effectively satisfies the ANC constraint but also promotes sparsity in the model.

The dual variable update includes the updates of three dual variables. According to (14), the update formula for $\mathbf{d}_1$ in the $(k+1)$th layer of the neural network is as follows:

$$\mathbf{d}_1^{k+1} = \mathbf{d}_1^k - \mathbf{W}_7\left(\mathbf{W}_6\mathbf{a}^{k+1} - \mathbf{v}_1^{k+1}\right) \tag{21}$$

[1]Here, any Mercer kernel is suitable. The adopted ACK satisfies Mercer's theorem.

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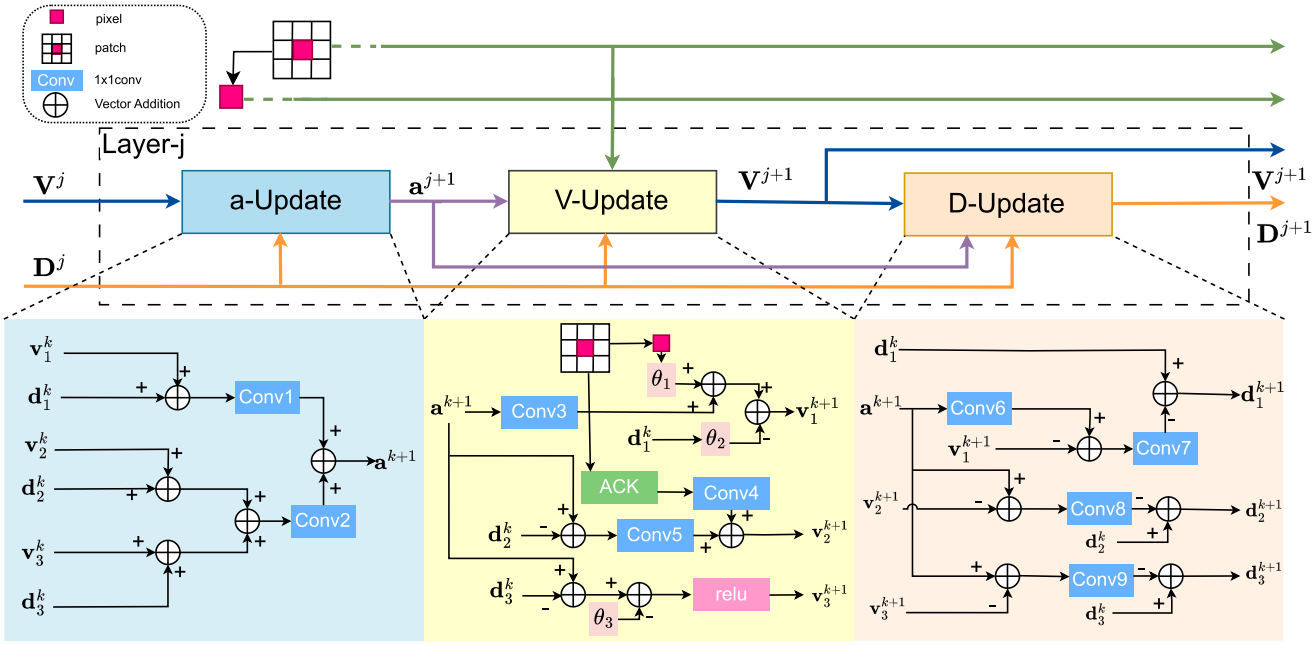


Fig. 2. Forward propagation process of **a**-update, **V**-update, and **D**-update layers in the network.

where $\mathbf{W}_6 \in \mathbb{R}^{L\times P}$ and $\mathbf{W}_7 \in \mathbb{R}^{L\times L}$ are learnable variables, achieved by 1×1 convolutions. $\mathbf{W}_6$ is initialized as $\mathbf{E}$, and $\mathbf{W}_7$ is a dynamic step size designed to adapt to the characteristics of the unmixing problem, enhancing the model's generalization ability.

Given that $\mathbf{d}_2$ and $\mathbf{d}_3$ possess identical update structures and solutions with the same form. According to (15) and (16), their update formulas in the $(k+1)$th layer of the neural network are as follows:

$$\begin{aligned}\mathbf{d}_2^{k+1} &= \mathbf{d}_2^k - \mathbf{W}_8\left(\mathbf{a}^{k+1} - \mathbf{v}_2^{k+1}\right)\\ \mathbf{d}_3^{k+1} &= \mathbf{d}_3^k - \mathbf{W}_9\left(\mathbf{a}^{k+1} - \mathbf{v}_3^{k+1}\right)\end{aligned} \tag{22}$$

where $\mathbf{W}_8 \in \mathbb{R}^{P\times P}$ and $\mathbf{W}_9 \in \mathbb{R}^{P\times P}$ are learnable variables, representing 1×1 convolutions. Like $\mathbf{W}_7$, $\mathbf{W}_8$ and $\mathbf{W}_9$ are also dynamic step sizes, better adapting to the unmixing problem.

Within the encoder, each layer processes interleaved inputs and outputs from the **a**-update, **V**-update, and **D**-update steps, with the structure of the layers reflecting a single iteration of the ADMM algorithm. In the last layer, $\mathbf{v}_3$ is normalized using the ℓ_1 norm to reintroduce the ASC constraint, and the final abundance is

$$\hat{\mathbf{a}} = \frac{\mathbf{v}_3^{k+1}}{\left\|\mathbf{v}_3^{k+1}\right\|_1}. \tag{23}$$

The encoder enhances interpretability by embedding the ADMM's structured optimization framework into neural networks, which transforms iterative optimization steps into transparent network layers. This integration not only clarifies the parameter update logic but also establishes a direct mathematical connection between model architecture and optimization principles, contrasting with the opaque nature of conventional deep learning approaches.

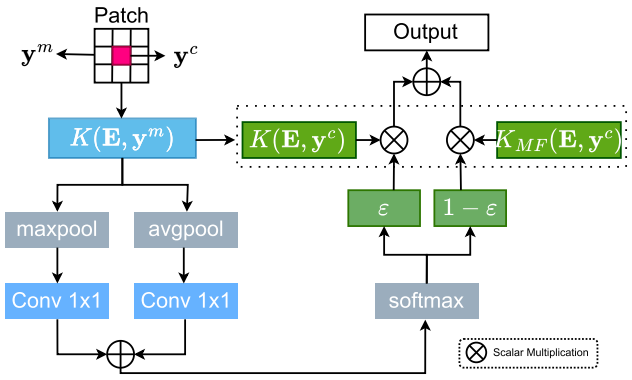


Fig. 3. Illustration of the ACK module.

B. ACK Module

To integrate the spatial information of HSIs into network, we design an ACK module (Fig. 3) that combines both SSK and traditional kernel.

SSK is initially employed for HSI classification tasks [64]. It is capable of capturing high-order similarities between neighboring samples within a complex manifold [67]. Consequently, we incorporate the SSK into our network architecture to facilitate the learning of spatial information. Given $\mathbf{y}$ and its neighbors $\mathbf{y}^m \in \Omega_y, m = 1, 2, \ldots, M^2$, where Ω_y is a spatial window with size M, in the proposed method, we utilize the mean value of the pixels within the window Ω_y subsequent to the application of the ϕ mapping function. The resultant vector is given as follows:

$$\mathrm{MF}(\phi(\mathbf{y})) = \frac{1}{M^2}\sum_{m=1}^{M^2}\phi(\mathbf{y}^m). \tag{24}$$

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Then, the kernel function K_{MF} is written as $K_{\mathrm{MF}}(\mathbf{y}_i, \mathbf{y}_j) = \langle \mathrm{MF}(\phi(\mathbf{y}_i)), \mathrm{MF}(\phi(\mathbf{y}_j)) \rangle$. However, given the absence of neighboring pixels in the endmembers, we employ an asymmetric form of the SSK

$$\begin{aligned} K_{\mathrm{MF}}(\mathbf{E}, \mathbf{y}_i) &= \langle \Phi(\mathbf{E}), \mathrm{MF}(\phi(\mathbf{y}_i)) \rangle \\ &= \left\langle \Phi(\mathbf{E}), \frac{1}{M^2} \sum_{m=1}^{M^2} \phi\left(\mathbf{y}_i^m\right) \right\rangle \\ &= \frac{1}{M^2} \sum_{m=1}^{M^2} K\left(\mathbf{E}, \mathbf{y}_i^m\right) \end{aligned} \tag{25}$$

where $K_{\mathrm{MF}}(\mathbf{E}, \mathbf{y}_i) \in \mathbb{R}^P$, which calculates the similarity between the neighbors of a pixel and each endmember.

In addition to using SSK to obtain spatial information, we also combine it with the normal kernel

$$K_{\mathrm{CK}}(\mathbf{E}, \mathbf{y}^c) = \varepsilon K_{\mathrm{MF}}(\mathbf{E}, \mathbf{y}^c) + (1 - \varepsilon) K(\mathbf{E}, \mathbf{y}^c) \tag{26}$$

where $\mathbf{y}^c$ is the center pixel of the input patch and $1 \geq \varepsilon \geq 0$ is to adjust the weight coefficient of the two kernel functions. A dynamic convolution-inspired approach [68] is introduced in the ACK, enabling the automatic adjustment of the parameter ε. This mechanism significantly enhances the model's capacity to adaptively capture intricate and diverse features, leading to improved performance in handling complex data patterns. The process begins with the computation of kernel functions between the patch and each endmember. Subsequently, the intermediate features obtained from this computation are processed through max-pooling and average-pooling layers. Specifically, average pooling assists the model in capturing global information by averaging the values across the entire region, thereby reducing noise and emphasizing overall patterns. On the other hand, max pooling captures salient local features by selecting the maximum value within the region, enabling the model to focus on the most important information. This pooling strategy allows the module to effectively extract both global and local features from the patch. Following this, a 1×1 convolution is applied to compress these features into two parameters. These parameters are then summed and activated using a softmax function to determine the coefficients for the two kernel functions.

Notably, unlike convolution-based methods constrained by linear filtering mechanisms, our ACK model establishes a dual adaptive system in reproducing kernel Hilbert space. While convolution relies on fixed-weight local operators that enforce translational symmetry, ACK dynamically combines SSK with traditional kernels through learnable weighting coefficients, enabling nonlinear similarity measurement between pixels and endmembers. Two fundamental advancements distinguish our approach: 1) the spectral-aware weighting mechanism adaptively adjusts kernel combinations based on global spectral characteristics, overcoming convolution's limitation in modeling cross-band correlations; and 2) spatial constraints embedded in the kernel alignment process capture both local pixel neighborhoods and global endmember distributions, a critical capability absent in standard convolutional networks.

C. Decoder

The decoder comprises two reconstruction branches: a linear branch and a nonlinear branch. The linear branch can be represented as

$$\hat{\mathbf{y}}^l = \mathrm{Linear}(\hat{\mathbf{a}}) = \hat{\mathbf{E}}^l \hat{\mathbf{a}} \tag{27}$$

where $\hat{\mathbf{y}}^l$ is the linear reconstructed pixel; $\hat{\mathbf{a}}$ is the estimated abundance vector; and $\hat{\mathbf{E}}^l = (\hat{\mathbf{e}}_1^l, \hat{\mathbf{e}}_2^l, \ldots, \hat{\mathbf{e}}_P^l) \in \mathbb{R}^{L \times P}$ is the weight, representing the endmembers of the linear part. The nonlinear part is derived from the nonlinear mapping (3)

$$\phi(\hat{\mathbf{y}}^n) = \Phi(\hat{\mathbf{E}}^n) \hat{\mathbf{a}} \tag{28}$$

where $\hat{\mathbf{y}}^n$ is the nonlinear reconstruction pixel and $\hat{\mathbf{E}}^n = (\hat{\mathbf{e}}_1^n, \hat{\mathbf{e}}_2^n, \ldots, \hat{\mathbf{e}}_P^n) \in \mathbb{R}^{L \times P}$ is the weight, representing the endmembers of the nonlinear part. Multiply both sides of (28) by $\phi^T(\mathbf{y})$, we get

$$K(\mathbf{y}, \hat{\mathbf{y}}^n) = K(\mathbf{y}, \hat{\mathbf{E}}^n) \hat{\mathbf{a}} \tag{29}$$

where $K(\mathbf{y}, \hat{\mathbf{E}}^n) = K^T(\hat{\mathbf{E}}^n, \mathbf{y})$ is implemented by the ACK module K_{CK} in Section III-B.

Notably, the two branches of the decoder correspond to the two fidelity terms of the biobjective problem (4). During the training process, the outputs of the two branches are dynamically balanced to ensure optimal performance. Specifically, while the linear decoder focuses on extracting endmembers that adhere to simpler linear relationships, its counterpart, the nonlinear decoder, captures more complex and high-dimensional features that remain inaccessible to linear methods. This collaborative approach enables the model to simultaneously leverage the strengths of both decoders through a weighted adjustment mechanism, which dynamically optimizes their relative contributions. The mechanism's weight adjustment is guided by the biobjective formulation, while the associated loss function quantitatively drives the update during optimization, thereby ensuring alignment of both linear and nonlinear objectives with the overall unmixing goal. To further enhance this process, the introduced ACK facilitates nonlinear relationship capture by learning pixel-neighborhood similarities. The final endmembers are derived from the linear branch; however, despite their primarily linear nature, the overall model also integrates nonlinear features, influenced by the loss function detailed in Section III-D. Moreover, both decoder branches share the same abundance vector $\mathbf{a}$, which serves as a consistent representation across the linear and nonlinear paths. This shared supervision helps to align the optimization objectives of both branches and stabilize the estimation of abundance during training.

D. Loss Function

The loss function is comprised of three terms. The linear term uses spectral angle distance (SAD), calculated as follows:

$$\mathcal{L}_{\mathrm{linear}} = \mathrm{SAD}(\mathbf{y}, \hat{\mathbf{y}}^l) = \arccos \frac{\langle \mathbf{y}, \hat{\mathbf{y}}^l \rangle}{\|\mathbf{y}\|_2 \left\|\hat{\mathbf{y}}^l\right\|_2} \tag{30}$$

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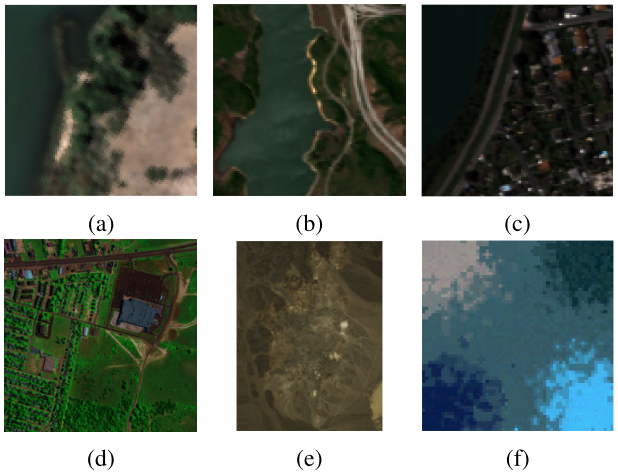


Fig. 4. Datasets used in the experiments. (a) Samson. (b) Jasper Ridge. (c) Apex. (d) Urban. (e) Cuprite. (f) Synthetic data.

where $\mathbf{y}$ represents the original pixel. The nonlinear term is calculated according to (29)

$$\begin{aligned}\mathcal{L}_{\text{nonlinear}} &= \left\| K_{\text{CK}}(\mathbf{y},\mathbf{y}) - K_{\text{CK}}(\mathbf{y},\hat{\mathbf{y}}^n) \right\|_1 \\ &= \| K_{\text{CK}}(\mathbf{y},\mathbf{y}) - K_{\text{CK}}(\mathbf{y},\hat{\mathbf{E}}^n)\hat{\mathbf{a}} \|_1. \end{aligned} \tag{31}$$

The last term is the similarity between the two kinds of endmembers

$$\begin{aligned}\mathcal{L}_{\text{endmember}} &= SAD(\hat{\mathbf{E}}^l, \hat{\mathbf{E}}^n) \\ &= \frac{1}{P}\sum_{i=1}^{P} \arccos \frac{\langle \hat{\mathbf{e}}_i{}^l, \hat{\mathbf{e}}_i{}^n \rangle}{\|\hat{\mathbf{e}}_i{}^l\|_2 \|\hat{\mathbf{e}}_i{}^n\|_2}. \end{aligned} \tag{32}$$

The total loss of the proposed model is defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{linear}} + \mathcal{L}_{\text{nonlinear}} + \delta \mathcal{L}_{\text{endmember}} \tag{33}$$

where $\delta > 0$ is a regularization parameter.

IV. Experiments

In this section, we evaluate the performance of the proposed model on five real datasets and one synthetic dataset by comparing it with 11 classic and state-of-the-art algorithms, including seven linear methods and four nonlinear methods. Additionally, we undertake ablation studies to quantify the individual contributions of each model component. Furthermore, we perform sensitivity analyses to examine the influence of various hyperparameters on the model's performance.

A. Datasets

1) Samson Dataset: The Samson dataset [69] was acquired using the SAMSON sensor and comprises 156 spectral bands spanning the 401–889-nm wavelength range. The dataset contains a 95 × 95 pixel image, which is a cropped section of a larger scene. The RGB image of Samson is shown in Fig. 4(a). The dataset is a scene composed of three main endmembers, commonly referred to as *Soil*, *Tree*, and *Water*.

2) Jasper Ridge Dataset: The Jasper Ridge dataset [70] was captured using the AVIRIS sensor, providing pixels with 224 spectral bands covering the 380–2500-nm wavelength range and offering a spectral interval of 9.45 nm. To minimize the effects of water vapor and atmospheric conditions, we perform atmospheric correction and exclude heavily affected bands. Afterward, we extract a 100 × 100 subimage containing 198 relatively unaffected bands from the original data. The RGB representation of the Jasper Ridge dataset is illustrated in Fig. 4(b). This dataset primarily consists of four endmembers, commonly identified as *road*, *water*, *soil*, and *tree*.

3) Apex Dataset: The Apex dataset was acquired using the APEX sensor [71], focusing on a wavelength range of 413–2420 nm. From this dataset, we extract a 110 × 110 pixel subimage. The RGB representation of the Apex scene is illustrated in Fig. 4(c). This dataset features a scene composed of four primary endmembers, commonly identified as *road*, *water*, *roof*, and *tree*.

4) Urban Dataset: The Urban dataset, with a resolution of 307 × 307 pixels, was acquired using the HYDICE sensor [72]. The data span 210 spectral bands from 400 to 2500 nm. In experiments, 48 bands were excluded due to interference arising from dense water vapor and atmospheric effects, leaving 162 bands for subsequent analysis. The RGB image associated with the dataset is shown in Fig. 4(d). This dataset consists of four main endmembers, i.e., *asphalt*, *grass*, *tree*, and *roof*.

5) Cuprite Dataset: The Cuprite dataset was captured using the AVIRIS sensor, and it possesses dimensions of 250 × 190 × 188 [13]. This dataset represents a complex scenario for mineral unmixing. The Cuprite scene primarily consists of 12 endmember minerals. Based on previous studies [73], the five most prominent minerals in this dataset are *Alunite*, *Buddingtonite*, *Chalcedony*, *Kaolin/Smect*, and *Montmorillonite*. Fig. 4(e) presents the corresponding RGB image.

6) Synthetic Dataset: To further verify the noise resistance of our model, we use the synthetic dataset, with a resolution of 60 × 60 pixels. This dataset selects five endmembers with 200 bands in the ASTER spectral library [74]. Additionally, the RGB image corresponding to the dataset is presented in Fig. 4(f). We add 10-, 20-, and 30-dB noise to the original image for experiments.

B. Experimental Setup and Performance Measures

The proposed model is evaluated against 11 methods. The seven linear unmixing methods are VCA-FCLS [15], LGSU [75], LSU [76], uDAS [39], CyCU-Net [41], UAD-MMNet [53], and A2SAN [48]. The four nonlinear unmixing methods are Model-based AutoenCoder (MACUNet) [56], HapkeCNN [57], 3DCNNAE [77], and MLM-3DAE [61]. Since LGSU and LSU are designed to address the problem of sparse regression unmixing, for fairness, we use the endmembers extracted by VCA as their spectral library.

Our model is implemented in Python 3.7 and PyTorch 1.12 with GPU (NVIDIA GeForce RTX 3060). In the encoder, the parameter α of the linear and nonlinear effects is set

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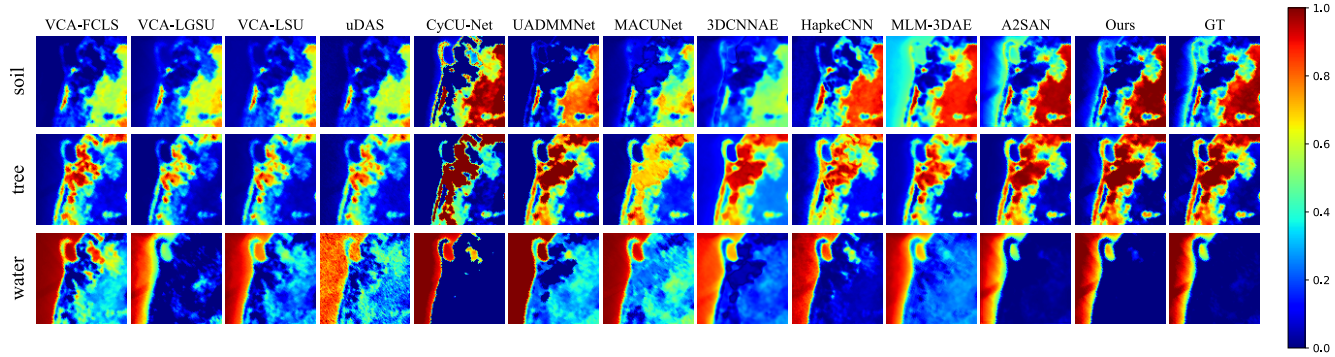


Fig. 5. Abundance RMSE comparison results on the Samson dataset.

TABLE I
PARAMETER SETTINGS OF THE PROPOSED NETWORK

Hyperparameters	Samson	Jasper Ridge	Apex	Urban	Cuprite	Synthetic
lr (Encoder)	1e - 4	1e - 4	1e - 4	1e - 5	1e - 5	5e - 4
lr (Decoder)	1e - 4	5e - 5	1e - 4	1e - 5	1e - 5	5e - 4
Weight Decay	1e - 3	1e - 5	1e - 5	1e - 4	1e - 8	1e - 4
γ	0.96	0.96	0.88	0.70	0.70	0.99

to 0.5. σ in the Gaussian kernel function is initialized to 2.0 and subsequently optimized concurrently with the network. The regularization parameter δ is assigned a value of 0.02. Additionally, λ, which controls the regularization strength for abundance sparsity in the optimization problem, is set to $1e^{-3}$, while μ, serving as a penalty parameter within the ADMM framework, is set to $1e^{-4}$. The initial endmember matrix $\mathbf{E}$ is extracted using the VCA method [13] and subsequently normalized to mitigate the effects of endmember variability. The patch size is determined to be 3, and the number of layers is set to 6, as elaborated in Section IV-E. The batch size is fixed at 512, with a maximum iteration limit of 300. After establishing the initial learning rate, this rate lr is reduced by a factor of γ every ten epochs. It is worth noting that weight decay is only applied in the decoder, as the complexity of the encoder is primarily managed by the ADMM algorithm, which inherently imposes constraints on the model. Due to the varying spectral characteristics and data distributions across different datasets, the learning rate lr, weight decay, and γ values are adjusted accordingly for each dataset, as detailed in Table I.

The quantitative results for abundance are evaluated by comparing the root mean squared error (RMSE) between the estimated results and the ground truth (GT). The formula is as follows:

$$\mathrm{RMSE}(\mathbf{A}, \hat{\mathbf{A}}) = 100 \times \sqrt{\frac{1}{N}\|\mathbf{A} - \hat{\mathbf{A}}\|_F^2} \tag{34}$$

where $\mathbf{A}$ and $\hat{\mathbf{A}}$ refer to the true and estimated abundances, respectively. The quantitative results for endmembers are evaluated by comparing the SAD between the estimated results and the GT. The formula is as follows:

$$\mathrm{SAD}(\mathbf{E}, \hat{\mathbf{E}}) = \frac{1}{P}\sum_{i=1}^{P} \arccos\left(\frac{\langle \mathbf{e}_i, \hat{\mathbf{e}}_i \rangle}{\|\mathbf{e}_i\|_2 \|\hat{\mathbf{e}}_i\|_2}\right)\frac{180}{\pi} \tag{35}$$

where $\mathbf{E}$ and $\hat{\mathbf{E}}$ refer to the true and estimated endmembers, respectively.

C. Experiment on Real Datasets

1) Samson: The quantitative results are summarized in Table II. The SAD results show that our proposed model performs consistently well across the *Soil*, *Tree*, and *Water* categories. While MACUNet displays notable accuracy in identifying *Tree*, our model attains results that are on par with MACUNet in this specific aspect. Additionally, our model demonstrates enhanced performance in the detection of other endmembers. Regarding the RMSE metric, our proposed model maintains good performance across the *Soil*, *Tree*, and *Water* categories. Although A2SAN remains comparable to our model in the *Soil* and *Tree* categories, our model exhibits a substantial advantage over A2SAN in the *Water* category.

Figs. 5 and 6 illustrate the abundance map and the endmember curves produced by the compared methods, respectively. As shown in Fig. 5, other models, including uDAS, CyCU-Net, UADMMNet, MACUNet, 3DCNNAE, and MLM-3DAE, demonstrate limited accuracy in estimating abundances for the Samson dataset. Notably, HapkeCNN struggles with the accurate estimation of *Tree* material, while A2SAN performs suboptimally in estimating *Water* content. In contrast, our proposed model provides superior abundance estimations for *Soil*, *Tree*, and *Water*, effectively capturing finer details. Furthermore, the endmember curves depicted in Fig. 6 reveal that HapkeCNN encounters difficulties in accurately extracting *Soil*, and most models fail to effectively capture *Water*. In contrast, the endmembers extracted by our proposed model closely align with the GT for all three materials.

2) Jasper Ridge: Table III presents the quantitative results for the Jasper Ridge dataset. Our proposed model exhibits consistent performance in the extraction of all four endmembers. While UADMMNet achieves a notable SAD value for the *Road* class, our model's performance is closely comparable. The average SAD value attained by our model further supports its overall effectiveness in endmember extraction. Regarding the RMSE metric, our model performs best in all classes.

Fig. 7 displays the abundance estimation maps produced by various models on the Jasper Ridge dataset. The results indicate that uDAS, CyCU-Net, and 3DCNNAE exhibit suboptimal performance in abundance estimation. Other unmixing

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TABLE II

SAD AND RMSE OF THE COMPARED METHODS WHEN APPLIED TO THE SAMSON DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods		VCA-FCLS	VCA-LGSU	VCA-LSU	uDAS	CyCU-Net	UADMMNet	MACUNet	3DCNNAE	HapkeCNN	MLM-3DAE	A2SAN	Ours
SAD	soil	1.18	—	—	1.91	2.17	1.25	2.46	1.21	4.09	2.34	0.63	**0.43**
	tree	2.83	—	—	3.27	2.08	1.95	**1.67**	2.86	5.09	3.73	2.55	1.85
	water	7.44	—	—	16.10	8.05	3.49	6.65	8.75	11.02	7.57	7.84	**1.65**
	mean	3.82	—	—	7.09	4.10	2.23	3.60	4.27	6.73	4.55	3.67	**1.31**
RMSE	soil	25.65	21.71	24.23	25.27	18.23	25.09	28.33	22.94	17.28	15.89	7.91	**7.17**
	tree	18.20	20.94	18.96	19.56	19.11	6.55	14.37	12.48	12.46	17.27	4.78	**4.64**
	water	34.88	8.09	21.99	33.82	15.66	27.49	29.66	15.88	12.56	20.95	6.37	**2.86**
	mean	27.12	18.03	21.84	26.87	17.73	21.48	25.09	17.64	14.28	18.16	6.48	**5.20**

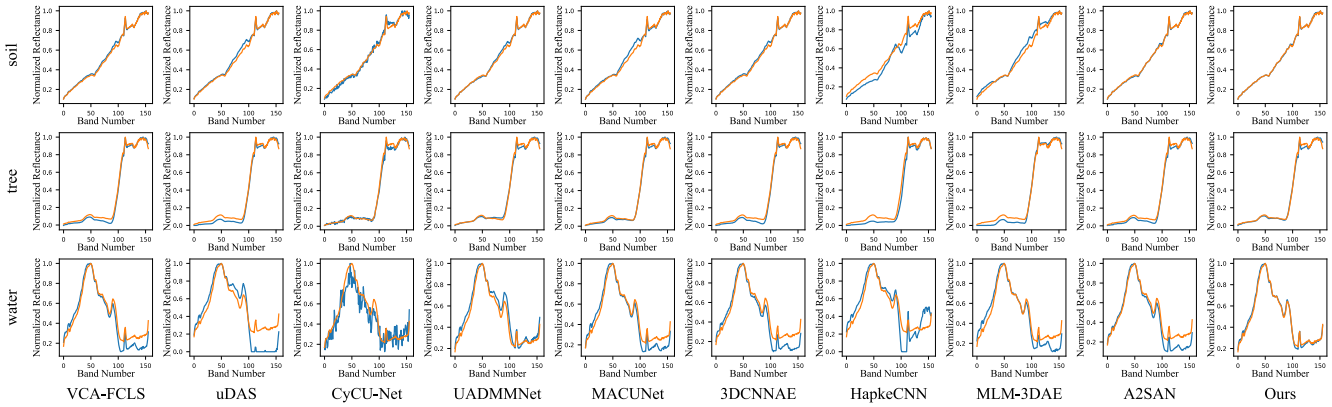


Fig. 6. Endmember SAD comparison results on the Samson dataset. Blue line: estimated endmembers. Orange line: reference endmembers.

TABLE III

SAD AND RMSE OF THE COMPARED METHODS WHEN APPLIED TO THE JASPER RIDGE. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods		VCA-FCLS	VCA-LGSU	VCA-LSU	uDAS	CyCU-Net	UADMMNet	MACUNet	3DCNNAE	HapkeCNN	MLM-3DAE	A2SAN	Ours
SAD	tree	6.13	—	—	7.39	3.09	8.31	7.97	6.36	13.18	6.98	6.77	**1.05**
	water	14.63	—	—	19.11	9.33	3.88	20.37	19.12	9.96	11.60	8.00	**1.62**
	dirt	6.68	—	—	7.15	3.93	7.57	6.50	6.14	9.99	8.03	3.61	**2.48**
	road	5.16	—	—	3.45	5.93	**2.66**	4.64	5.97	6.36	4.05	7.43	2.83
	mean	8.15	—	—	9.27	5.58	5.60	9.87	9.40	9.87	7.67	6.45	**1.99**
RMSE	tree	15.94	15.96	15.78	16.32	10.04	15.29	17.50	14.77	16.88	13.88	11.27	**3.48**
	water	21.64	26.55	23.67	20.81	10.16	17.11	18.30	19.27	3.93	16.23	10.34	**3.00**
	dirt	13.54	13.74	13.93	13.60	14.53	8.09	12.52	17.87	13.33	13.99	12.55	**5.69**
	road	10.80	10.73	10.26	11.27	11.76	12.15	11.75	26.10	18.77	12.94	10.64	**4.50**
	mean	15.99	17.77	16.65	15.90	11.76	13.60	15.30	19.94	14.41	14.31	11.23	**4.29**

techniques also face challenges in accurately estimating the abundance of certain materials. For example, MACUnet experiences challenges in estimating *Tree* abundance, while HapkeCNN, MLM-3DAE, and A2SAN demonstrate limited accuracy in assessing *Road* abundance. In contrast, the abundance map generated by our model closely aligns with the GT. Fig. 8 presents the endmember curves extracted by different unmixing techniques on the Jasper Ridge dataset. The findings suggest that uDAS, MACUNet, and 3DCNNAE are ineffective at extracting the *Water* class, while HapkeCNN falls short in extracting the *Tree* class. Although UADMMNet produces a *Road* endmember curve that closely resembles the true endmember, its performance in extracting other endmembers is slightly inferior to that of our proposed model.

3) Apex: Table IV shows the quantitative results of endmembers extracted and abundance estimated by each unmixing model on the Apex dataset. With respect to the SAD metric, uDAS, 3DCNNAE, and A2SAN achieve the best results in extracting *Road*, *Tree*, and *Roof* classes, respectively. Additionally, uDAS and A2SAN achieve the second-best results in the task of extracting the *Roof* and *water* classes. However, their performance for other endmembers is suboptimal, resulting in an elevated overall mean SAD. Although our model does not achieve the best results for *Road*, *Tree*, and

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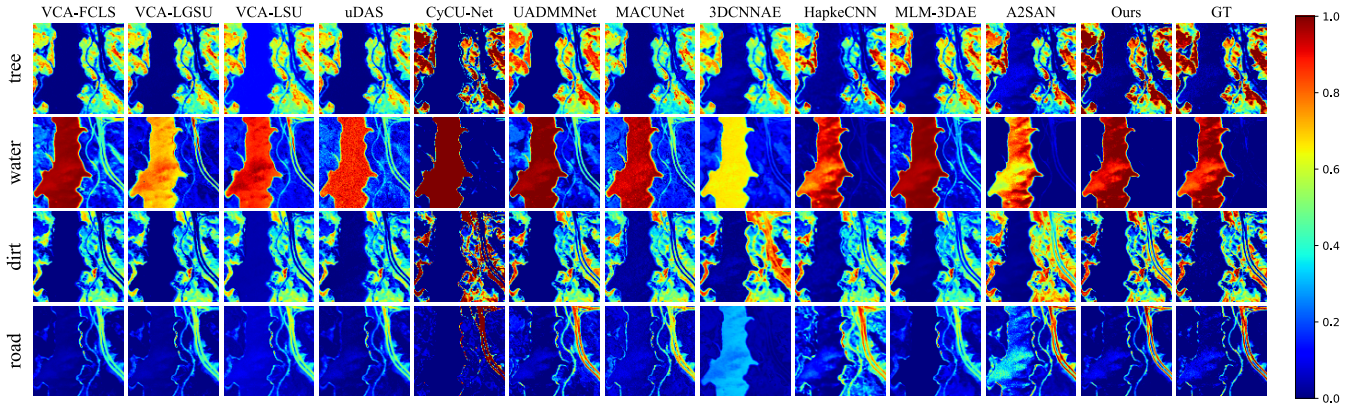


Fig. 7. Abundance RMSE comparison results on the Jasper Ridge dataset.

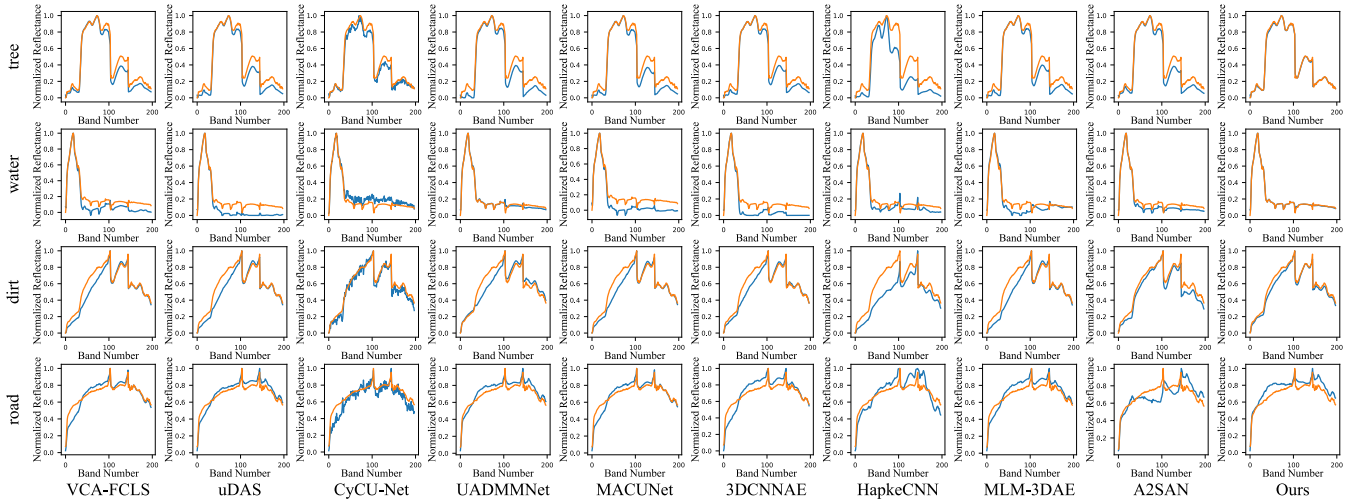


Fig. 8. Endmember SAD comparison results on the Jasper ridge dataset. Blue line: estimated endmembers. Orange line: reference endmembers.

TABLE IV

SAD AND RMSE OF THE COMPARED METHODS WHEN APPLIED TO THE APEX. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods		VCA-FCLS	VCA-LGSU	VCA-LSU	uDAS	CyCU-Net	UADMMNet	MACUNet	3DCNNAE	HapkeCNN	MLM-3DAE	A2SAN	Ours
SAD	road	5.66	—	—	**5.13**	14.89	6.41	7.14	5.72	6.12	6.35	16.13	5.18
	tree	5.49	—	—	6.63	7.00	6.96	7.99	**5.47**	10.34	5.84	7.72	7.17
	roof	15.11	—	—	9.00	11.91	11.93	11.97	15.10	9.79	13.49	**4.73**	9.55
	water	10.99	—	—	9.99	40.88	10.33	22.72	11.01	33.65	15.65	2.76	**2.31**
	mean	9.31	—	—	7.68	18.67	8.91	12.46	9.33	14.98	10.33	7.83	**6.05**
RMSE	road	17.57	17.81	17.98	17.62	21.39	18.15	17.89	18.19	22.95	19.97	**13.28**	16.18
	tree	24.19	24.00	24.24	24.12	28.67	20.10	22.88	16.76	18.24	14.76	13.69	**11.76**
	roof	22.66	22.57	22.05	22.89	24.24	24.41	20.43	20.38	16.04	20.33	11.52	**10.41**
	water	41.47	84.77	46.64	41.34	25.99	38.02	32.95	21.69	**10.26**	27.33	17.05	13.21
	mean	27.96	46.34	29.88	27.95	25.21	26.34	24.22	19.35	17.48	21.08	14.03	**13.07**

Roof individually, it outperforms other unmixing techniques in terms of the mean SAD. The RMSE metric serves as an indicator of the performance of the abundance estimation results. While our model's results for *Road* and *Water* abundance are not the best, they are closely comparable to the top-performing models. Overall, our model demonstrates superior performance in terms of the mean RMSE for abundance estimation compared to other techniques.

The visualization of the abundance maps for the Apex dataset is presented in Fig. 9. The results indicate that the abundance estimations produced by uDAS, CyCU-Net, UADMMNet, and MACUNet are suboptimal across all materials.

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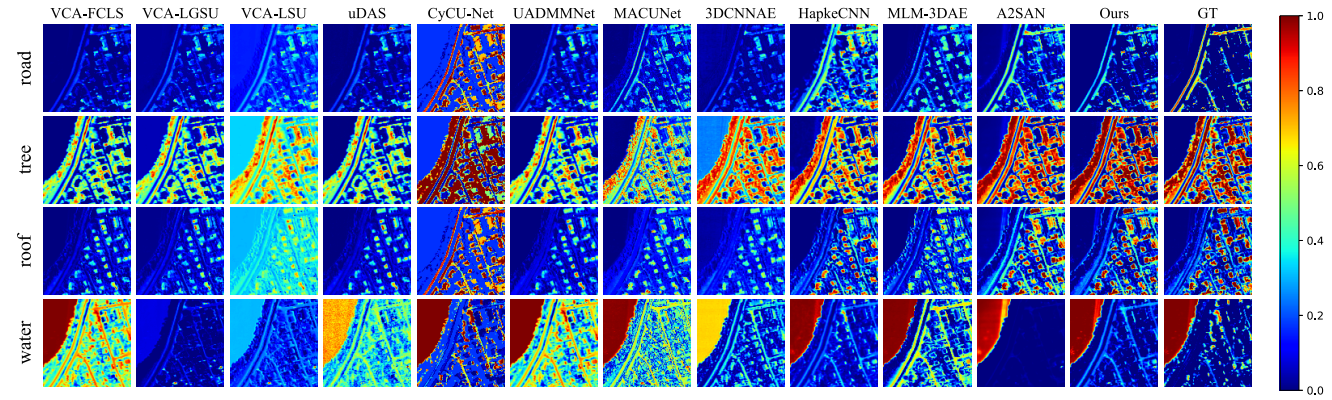


Fig. 9. Abundance RMSE comparison results on the Apex dataset.

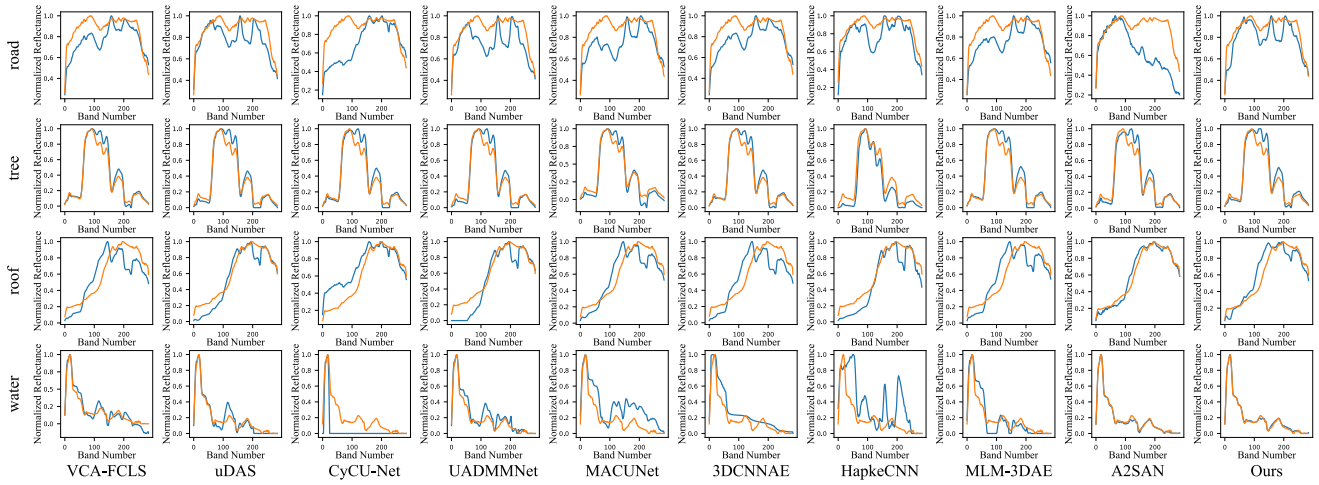


Fig. 10. Endmember SAD comparison results on the Apex dataset. Blue line: estimated endmembers. Orange line: reference endmembers.

TABLE V

SAD AND RMSE OF THE COMPARED METHODS WHEN APPLIED TO THE URBAN. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods		VCA-FCLS	VCA-LGSU	VCA-LSU	uDAS	CyCU-Net	UADMMNet	MACUNet	3DCNNAE	HapkeCNN	MLM-3DAE	A2SAN	Ours
SAD	asphalt	7.62	—	—	9.45	7.11	11.30	12.00	8.85	14.61	6.66	7.69	**6.40**
	grass	32.88	—	—	6.92	**2.63**	9.97	10.73	9.21	13.70	19.59	6.66	4.08
	tree	3.19	—	—	9.13	3.55	6.87	27.65	5.15	9.82	7.24	5.35	**1.64**
	roof	9.60	—	—	13.64	23.72	**2.59**	21.74	16.78	13.61	9.70	18.90	9.76
	mean	13.32	—	—	9.79	9.25	7.68	18.03	10.00	12.93	10.80	9.65	**5.47**
RMSE	asphalt	30.78	31.28	31.87	38.43	40.08	23.97	25.39	33.40	35.42	34.09	**11.27**	14.39
	grass	50.69	21.20	17.18	61.96	38.05	31.41	26.05	20.24	48.68	30.89	**12.31**	14.80
	tree	29.95	26.78	27.65	36.14	19.65	16.77	25.22	25.23	23.92	22.72	9.64	**7.41**
	roof	19.72	18.28	18.03	21.11	20.75	16.69	18.56	23.63	18.66	15.90	9.09	**6.97**
	mean	34.65	24.90	24.50	42.03	31.11	23.03	24.00	26.07	33.71	26.86	**10.66**	11.51

Furthermore, the abundance maps generated by HapkeCNN and MLM-3DAE demonstrate insufficient sparsity, leading to inaccurate estimations. In contrast, A2SAN exhibits excessively sparse *Water* abundance, resulting in suboptimal overall performance. Our proposed model, however, achieves a balanced level of sparsity, yielding superior results compared to the other models. Fig. 10 illustrates the endmember curves extracted by each model for the Apex dataset. Although our model exhibits certain discrepancies in the extraction of *Tree* endmembers when compared to the GT, it nonetheless demonstrates a closer alignment with the GT than other models that identify a larger number of endmembers.

4) Urban: Table V presents the quantitative results of endmember extraction and abundance estimation for each

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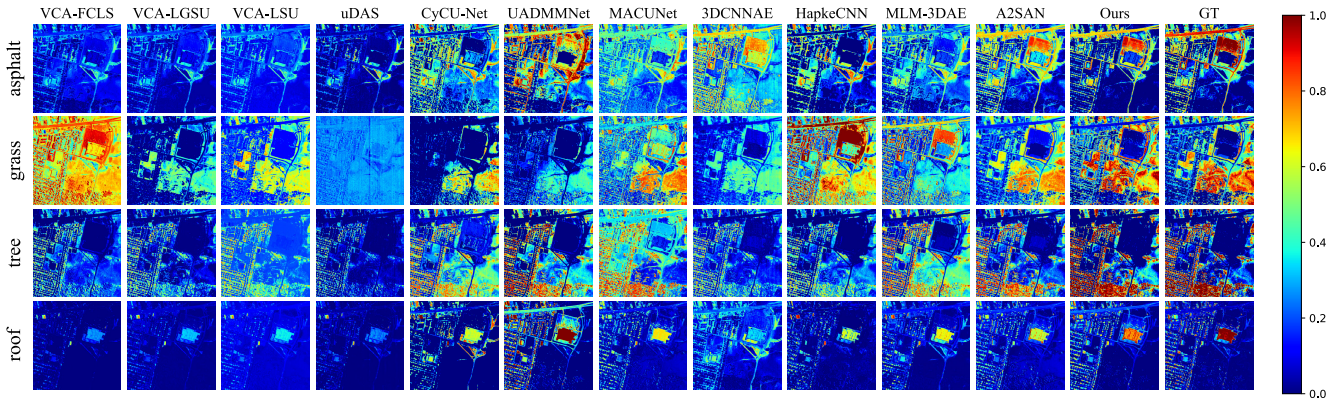


Fig. 11. Abundance RMSE comparison results on the Urban dataset.

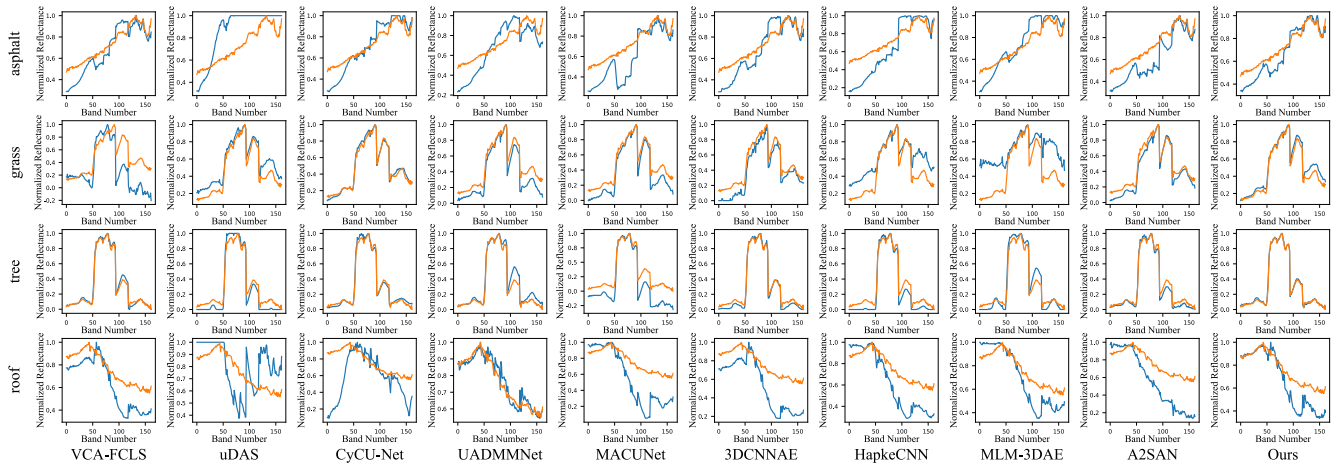


Fig. 12. Endmember SAD comparison results on the Urban dataset. Blue line: estimated endmembers. Orange line: reference endmembers.

unmixing method applied to the Urban dataset. In terms of the SAD metric, CyCU-Net demonstrates superior performance in extracting the *grass* class, while UADMMNet achieves the best results for the *roof* class. Although our method does not excel in individual categories, it outperforms other methods in the overall average SAD. Regarding the RMSE metric for abundance estimation, our method achieves the best results for the *tree* and *roof* classes. While we may not achieve the lowest average RMSE, our results are close to the best. Overall, our method performs well in both SAD and RMSE metrics, indicating significant potential in HU.

Fig. 11 visualizes the abundance maps of the Urban dataset. Most models struggle to accurately resolve the abundances of this dataset, possibly due to difficulties in capturing the actual sparsity level, leading to inaccurate abundance estimation. In contrast, our model adapts well to the sparsity of this dataset, resulting in visually superior outcomes. Fig. 12 displays the endmember curves for each model. Our model shows some bias in the *asphalt* and *roof* classes but fits well with the other two classes, demonstrating good overall visual performance.

5) Cuprite: Table VI shows the quantitative results of endmembers extracted by each unmixing model on the Cuprite dataset. Given the absence of GT abundance maps in this dataset, the quantitative assessments of abundance estimations provided by each unmixing model have been excluded from the table. It can be seen that our model achieves the closest average SAD to the GT among the compared blind unmixing models. Fig. 13 presents the abundance maps of our model for five key minerals, *Alunite*, *Buddingtonite*, *Chalcedony*, *Kaolin/Smect*, and *Montmorillonite*, using LSU results as a reference benchmark. From Fig. 13, it can be observed that, owing to the inherent uncertainties associated with endmember

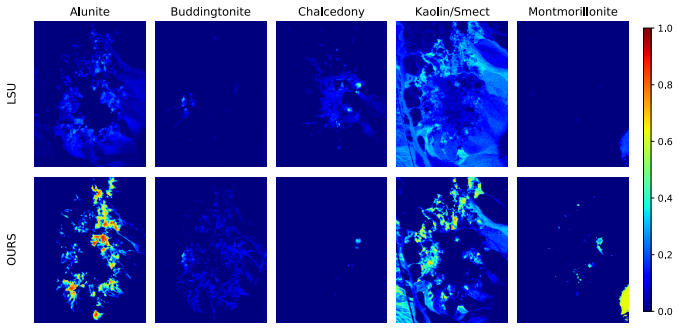


Fig. 13. Fractional abundance maps estimated for the Cuprite image.

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TABLE VI

SAD OF THE COMPARED METHODS WHEN APPLIED TO THE CUPRITE. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods	VCA	uDAS	CyCU-Net	UADMMNet	MACUNet	3DCNNAE	HapkeCNN	MLM-3DAE	A2SAN	Ours
Mean SAD	7.20	7.56	8.31	8.47	6.56	24.42	6.97	6.24	7.41	**5.07**

TABLE VII

COMPARISON OF RUNTIMES IN SECONDS (S) FOR DIFFERENT METHODS. THE BOLD ONES ARE THE BEST RESULTS

Methods	VCA-FCLS	LGSU	LSU	uDAS	CyCU-Net	UADMMNet	MACUNet	3DCNNAE	HapkeCNN	MLM-3DAE	A2SAN	Ours
Samson	2.89	14.94	**0.54**	2.71	1.48	1.20	69.40	2.29	2.39	2.23	1.58	1.63
Jasper Ridge	4.82	18.27	**0.56**	10.88	2.27	1.37	75.96	2.33	2.61	2.66	1.63	1.68
Apex	5.68	23.49	**0.51**	20.35	2.94	1.87	114.25	3.04	2.95	2.81	1.66	2.11
Urban	46.76	524.41	**1.79**	825.47	5.04	3.17	381.39	13.13	9.60	6.02	1.95	8.54

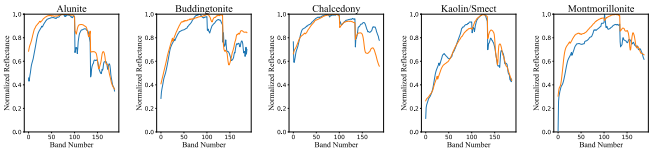


Fig. 14. Endmember SAD comparison results on the Cuprite dataset by the proposed model. Blue line: estimated endmembers. Orange line: reference endmembers.

identification in blind unmixing, our model displays certain discrepancies in finer details when compared to the LSU method. Nevertheless, on the whole, the results produced by our model exhibit a close correspondence with the reference, thereby underscoring the potential of our approach in tackling complex unmixing scenarios. In addition, Fig. 14 shows the corresponding endmember curves comparison between the proposed model and the reference curves, where the proposed model captures the main spectral characteristics of the key minerals with reasonable accuracy, further confirming its capability in accurately capturing endmember characteristics.

6) Running Time: In practical applications, we focus primarily on runtime rather than training time. Table VII presents the runtime of different methods on real datasets. Note that for LGSU and LSU, we only report the time required for abundance estimation. Among these methods, LSU achieves the fastest runtime due to its multilayer cascade approach, where each iteration progressively refines the solution toward optimality. In contrast, traditional deep learning models such as 3DCNNAE often have significantly higher runtime due to their complex architectures. Our method maintains a good balance between computational cost and performance, making it a practical and efficient choice.

D. Experiment on Synthetic Datasets

Table VIII summarizes the mean SAD and RMSE for the synthetic datasets across different noise levels (10, 20, and 30 dB) for the comparison models. After conducting a comprehensive analysis, it is observed that CyCU-Net and 3DCNNAE demonstrate suboptimal performance across all evaluated noise levels. Although MACUNet achieves favorable endmember results, attributed to its VCA initialization, the effectiveness of its abundance estimation is found to be relatively limited. In comparison, UADMMNet exhibits comparable performance to our proposed model at the 30-dB noise level; however, its performance suffers a notable decline when subjected to higher noise levels, specifically at 10 and 20 dB. While A2SAN demonstrates considerable robustness against noise, our model consistently outperforms it in terms of the overall SAD and RMSE metrics.

Figs. 15 and 16 present the endmember curves and abundance maps generated by each model at a noise level of 20 dB, respectively. The endmember curves generated by CyCU-Net and 3DCNNAE show a less satisfactory fit to the true endmembers. In contrast, the other models produce curves that more closely align with the actual endmembers. Nonetheless, an examination of the abundance maps reveals that our proposed model exhibits superior sparsity capability, resulting in a distribution that more accurately reflects the true abundances.

E. Ablation Experiment

1) Different Kernel Functions: Our proposed model integrates kernel tricks within both the encoder and decoder. In addition to the Gaussian kernel, we employ two other kernels, a polynomial kernel $K(\mathbf{x}, \mathbf{y}) = (\langle \mathbf{x}, \mathbf{y} \rangle + c_1)^d$ with parameters $d = 2$ and $c_1 = 1$, and a sigmoid kernel $K(\mathbf{x}, \mathbf{y}) = \tanh(\beta \langle \mathbf{x}, \mathbf{y} \rangle + c_2)$ with parameters $\beta = 0.01$ and $c_2 = 1$, both of which are optimized alongside the network. To assess their impact, we conduct a series of ablation experiments utilizing them, with the detailed results presented in Table IX. The results indicate that the Gaussian kernel function significantly outperforms the other kernel functions examined. The Gaussian kernel offers the advantage of mapping input data into an infinite-dimensional feature space, enabling it to capture complex patterns and nonlinear relationships within the data effectively. Furthermore, the utilization of the Gaussian kernel enables the construction of smoother decision boundaries within the feature space, thereby aiding in the mitigation of the risk of overfitting.

2) Different Kernel Modules: Table X presents the results obtained from real datasets utilizing the traditional kernel, the SSK, and the proposed ACK. While the traditional kernel achieves the lowest RMSE on the Samson dataset, the ACK gives suboptimal results and demonstrates superior perfor-

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TABLE VIII

SAD AND RMSE OF THE COMPARED METHODS WHEN APPLIED TO THE SYNTHETIC DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BLUE

Methods		VCA-FCLS	VCA-LGSU	VCA-LSU	uDAS	CyCU-Net	UADMMNet	MACUNet	3DCNNAE	HapkeCNN	MLM-3DAE	A2SAN	Ours
SAD	10dB	3.80	—	—	4.00	25.57	3.52	3.73	18.74	3.69	3.00	1.98	**1.18**
	20dB	1.73	—	—	1.72	25.45	1.40	1.70	16.30	1.83	1.69	1.91	**0.56**
	30dB	1.16	—	—	1.21	25.21	0.51	1.13	16.06	1.65	1.27	1.87	**0.46**
RMSE	10dB	11.29	7.94	11.49	13.81	27.61	13.83	28.74	24.25	17.18	11.09	8.85	**6.16**
	20dB	6.59	5.43	5.53	8.37	27.34	8.65	27.83	13.86	9.89	7.22	7.45	**5.24**
	30dB	5.38	5.10	5.12	6.65	27.30	5.22	28.03	16.14	8.41	6.88	7.06	**4.99**

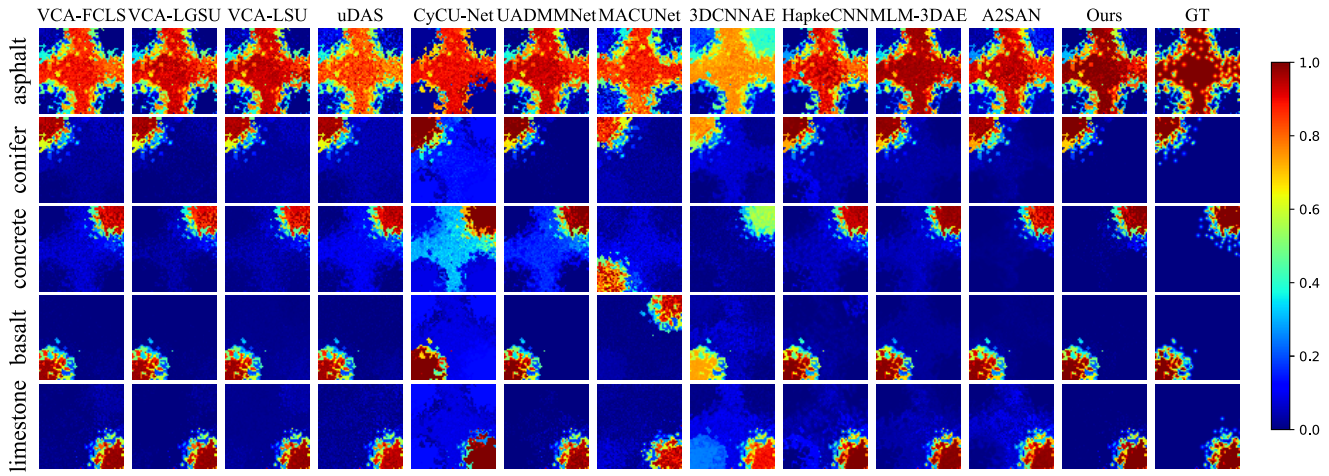


Fig. 15. Abundance RMSE comparison results on the synthetic dataset (20 dB).

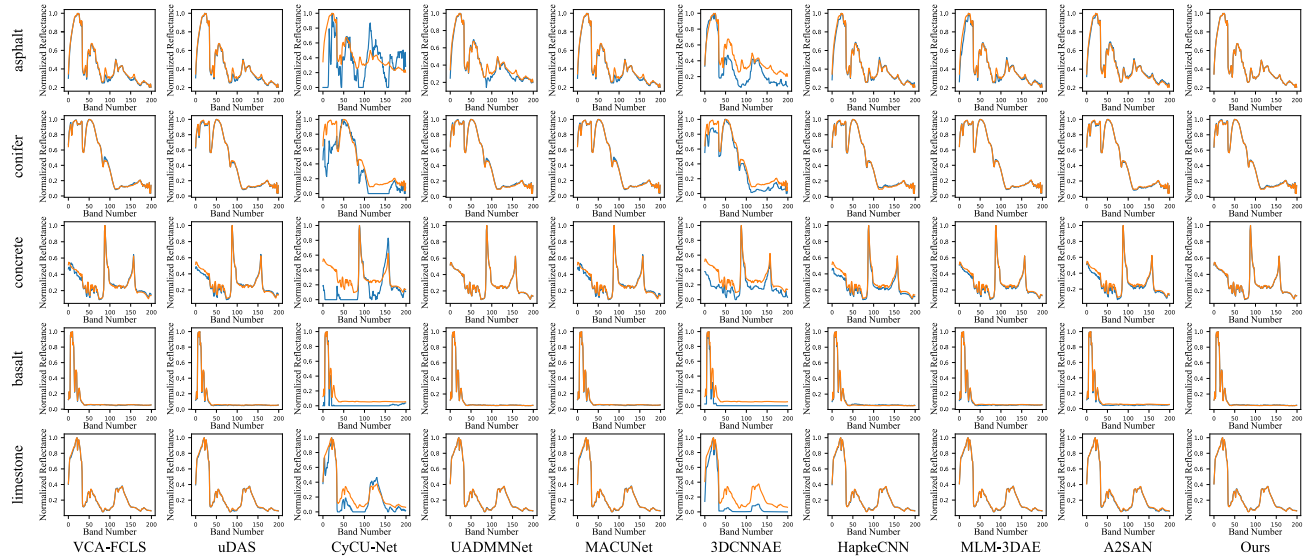


Fig. 16. Endmember SAD comparison results on the synthetic dataset (20 dB). Blue line: estimated endmembers. Orange line: reference endmembers.

mance across the remaining metrics. By incorporating spatial information and refining the central pixel data, the SSK and ACK achieve enhanced unmixing results. Unlike traditional kernels, the ACK combines two kernel functions and dynamically adjusts their weights based on the spatial–spectral relationships within the hyperspectral data. This adaptability allows the ACK to capture more complex and varying patterns in the data, leading to more accurate endmember extraction and abundance estimation. Furthermore, the flexible integration of both spatial and spectral information enables the ACK to better handle local variations and global trends in HSI, aspects that traditional kernels may struggle to model effectively.

3) Linear and Nonlinear Decoders: Table XI presents the results for different decoder configurations, revealing that the combination of linear and nonlinear decoders outperforms

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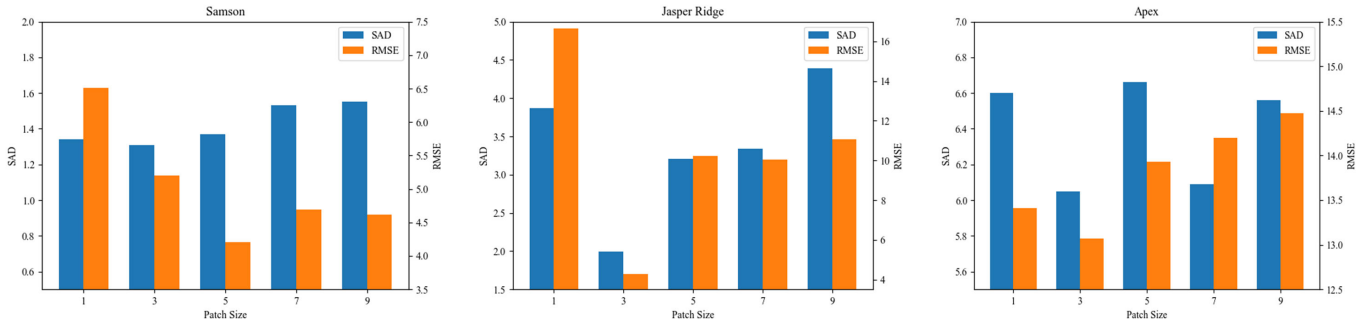


Fig. 17. Effect of varying patch sizes on SAD and RMSE for real datasets.

TABLE IX
ABLATION EXPERIMENTS OF DIFFERENT KERNEL FUNCTIONS ON DIFFERENT REAL DATASETS. THE BOLD ONES ARE THE BEST RESULTS

Dataset	Samson		Jasper Ridge		Apex	
Kernels	SAD	RMSE	SAD	RMSE	SAD	RMSE
Polynomial	4.15	11.72	8.71	10.69	6.74	14.74
Sigmoid	3.79	5.42	7.78	8.51	7.03	14.40
Gaussian	**1.31**	**5.20**	**1.99**	**4.29**	**6.05**	**13.07**

TABLE X
ABLATION EXPERIMENTS OF DIFFERENT KERNEL MODULES ON DIFFERENT REAL DATASETS. THE BOLD ONES ARE THE BEST RESULTS

Dataset	Samson		Jasper Ridge		Apex	
Kernel Modules	SAD	RMSE	SAD	RMSE	SAD	RMSE
Traditional Kernel	1.48	**4.39**	3.86	16.69	6.75	14.17
SSK	1.60	4.73	2.27	4.48	6.07	13.57
ACK	**1.31**	5.20	**1.99**	**4.29**	**6.05**	**13.07**

TABLE XI
ABLATION EXPERIMENTS OF LINEAR AND NONLINEAR DECODERS ON DIFFERENT REAL DATASETS. THE BOLD ONES ARE THE BEST RESULTS

Dataset	Samson		Jasper Ridge		Apex	
Decoder	SAD	RMSE	SAD	RMSE	SAD	RMSE
Linear	4.90	7.65	9.28	10.07	7.02	14.16
Nonliear	16.37	26.58	7.82	20.47	8.77	26.35
Linear+Nonliear	**1.31**	**5.20**	**1.99**	**4.29**	**6.05**	**13.07**

the use of a single decoder. This improvement is attributed to the fact that real-world datasets exhibit both linear and nonlinear characteristics. By employing both types of decoders simultaneously, the model can more effectively balance and integrate linear and nonlinear effects in conjunction with the encoder, leading to enhanced overall performance.

4) Endmember Initialization Strategies: Table XII presents the results obtained using different initialization strategies, including VCA, NMF, and random initialization. Among them, VCA typically yields strong performance, which can be attributed to its underlying geometric interpretation. Nevertheless, our model remains effective even without assuming the presence of pure pixels during initialization. This result highlights the model's ability to progressively refine endmember representations during training, thereby mitigating dependence on the pure-pixel assumption. Overall, the comparison underscores the robustness of the proposed model across various initialization strategies.

TABLE XII
ABLATION EXPERIMENTS OF DIFFERENT ENDMEMBER INITIALIZATION STRATEGIES ON REAL DATASETS. THE BOLD ONES ARE THE BEST RESULTS

Dataset	Samson		Jasper Ridge		Apex	
Initialization	SAD	RMSE	SAD	RMSE	SAD	RMSE
Random	1.91	**4.99**	3.92	12.78	8.60	13.36
NMF	1.47	6.01	2.48	7.37	**5.59**	15.23
VCA	**1.31**	5.20	**1.99**	**4.29**	6.05	**13.07**

F. Sensitivity Analysis of Model Parameters

1) Patch Size: Fig. 17 illustrates the impact of different patch sizes on unmixing performance for real datasets. With the exception of the RMSE for the Samson dataset, the results indicate that a patch size of 3 achieves optimal performance across the other metrics. A patch size of 1 fails to capture spatial information effectively, leading to an overemphasis on the central pixel. Conversely, an excessively large patch size incorporates redundant information from surrounding pixels, which ultimately degrades the overall results.

2) Different Numbers of Layers: Fig. 18 analyzes the relationship between the accuracy of the proposed model and the number of layers. Generally, an increase in the number of layers is correlated with an improvement in unmixing performance. However, an excessively high number of layers can result in a decrease in overall effectiveness. Specifically, a model with too many layers incurs significant computational costs, potential information loss, and slower convergence rates. Conversely, a model with too few layers may be inadequate in capturing the complex characteristics of the data. Therefore, it has been determined that a total of six layers provides the optimal balance for our model.

3) Different Values of α: Fig. 19 explores the relationship between model accuracy and α. Generally, adjusting α within a moderate range helps balance the linear and nonlinear effects of the model. However, an excessively large α reduces the influence of nonlinear components, leading to underutilization of nonlinear features and a decline in accuracy. Conversely,

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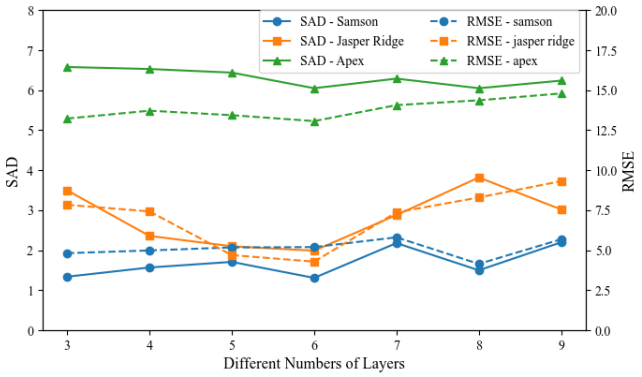


Fig. 18. Effect of different numbers of layers on SAD and RMSE for real datasets.

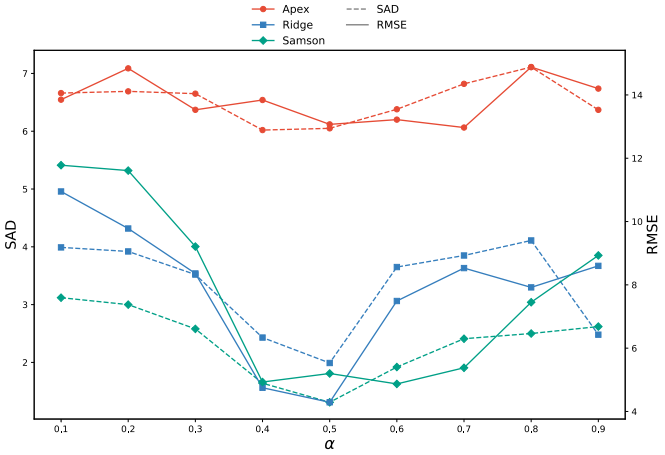


Fig. 19. Impact of α on SAD and RMSE for real datasets.

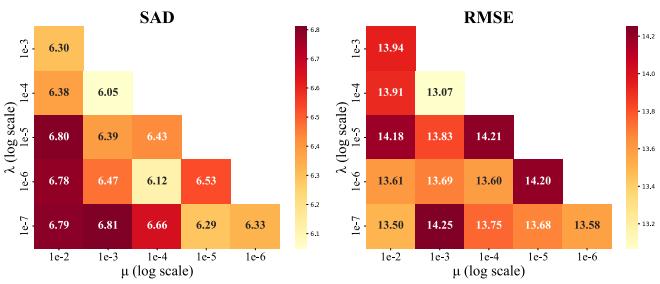


Fig. 20. Impact of λ and μ on SAD and RMSE for Apex datasets.

a very small α overemphasizes nonlinear effects, diminishing the contribution of linear components. Empirical evaluations indicate that the optimal range is $\alpha \in [0.4, 0.6]$, yielding improved unmixing performance.

4) Different Values of λ and μ: Fig. 20 investigates the impact of the regularization parameter λ and the penalty parameter μ. When the ratio λ/μ is greater than 1, the training process falls into an ill-conditioned optimization scenario. In this situation, a larger λ causes the ℓ_1 sparsity constraint to dominate the objective function, excessively shrinking the elements of the abundance vector to zero, while a smaller μ fails to effectively maintain the Lagrangian dual relationship of the residual constraint through the penalty term. Consequently,

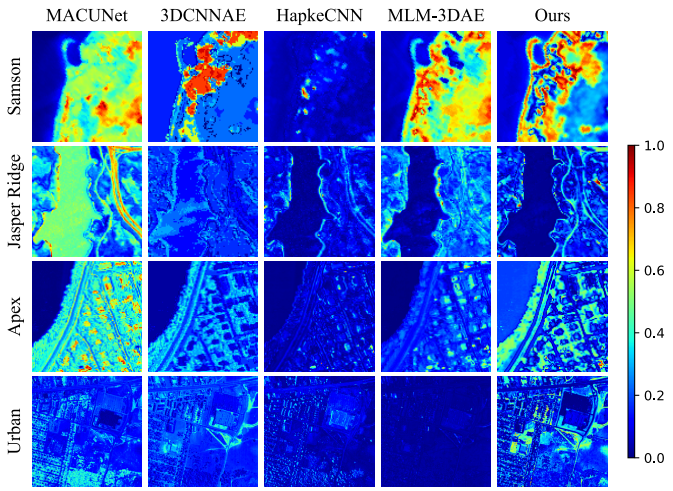


Fig. 21. Comparison of estimated nonlinear intensity maps for the Samson, Jasper Ridge, Apex, and Urban datasets.

the endmember learning becomes trapped in a local minimum, ultimately leading to physically uninterpretable updates of network parameters and an inability to converge to a meaningful solution space. Therefore, we focus on the case where $\lambda/\mu < 1$. In the figure, it is observed that an optimal sparsity level is attained when $\lambda/\mu = 0.1$. However, if λ/μ is excessively low, the sparsity becomes inadequate, degrading the unmixing performance. The empirical results show that the best performance is achieved when $\lambda = 1e^{-4}$ and $\mu = 1e^{-3}$.

G. Discussion

The aforementioned experimental results validate the effectiveness of our proposed method for blind HU. To further highlight the spatial distribution of nonlinear effects, Fig. 21 presents the estimated nonlinear intensity maps for the Samson, Jasper Ridge, Apex, and Urban datasets. The nonlinear intensity maps are generated by computing the absolute difference between reconstructed HSIs produced by the full model (including the nonlinear module) and those produced by a reduced model (excluding the nonlinear module and related loss terms). These absolute differences are then averaged across all spectral bands to yield a single intensity value per pixel, and the resulting map is normalized for visualization purposes.

It can be seen that these effects are not uniformly distributed but are concentrated in specific regions due to terrain variations, material heterogeneity, and multiple scattering effects. Our method effectively captures these complex nonlinearities while maintaining the interpretability of the linear branch. For instance, in the Samson dataset, linear methods struggle with the RMSE values for the *soil* and *tree* classes. Our model, however, reasonably infers the nonlinear intensity in these regions, thereby mitigating this challenge. Similarly, the nonlinear intensity maps for the Jasper Ridge, Apex, and Urban datasets reveal significant nonlinear interactions in highly mixed regions. Notably, in the Jasper Ridge dataset, although the dominant nonlinear interactions occur in mixed terrain regions, a moderate level of nonlinear intensity is also

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observed in the *tree* areas. Likewise, in the Apex dataset, the *tree* and *roof* regions exhibit prominent nonlinear effects, likely due to material composition variations and multiple scattering phenomena. These findings further highlight the advantages of our adaptive kernel framework over fixed nonlinear models like MLM-3DAE, which may have limitations in capturing dataset-specific nonlinearities, as observed in the Urban dataset. Compared to physics-based models like HapkeCNN, which rely on predefined physical assumptions, our approach generalizes better to unknown nonlinearities while preserving the physical interpretability of the model.

V. Conclusion

In this article, we propose a novel biobjective autoencoder-based deep learning method for blind unmixing of HSIs. Our approach integrates both linear and nonlinear components to address the complexities inherent in HU. Specifically, the encoder is structured by unrolling the ADMM algorithm, enhancing the model's interpretability. The decoder, tailored to tackle both linear and nonlinear unmixing simultaneously, independently learns the endmembers for each objective and updates them synchronously via the loss function. To effectively incorporate spatial information, we introduce the ACK within both the encoder and the decoder. This ACK seamlessly integrates SSK and traditional kernel, dynamically adjusting their respective weights during the computation to optimize the extraction of spatial features. Comparative experiments demonstrate that our model outperforms other models in estimating both abundance and endmembers, showcasing its ability to balance robustness and precision. However, despite these advancements, spectral variability remains a considerable challenge that our model does not fully address. Furthermore, the varying contributions of linear and nonlinear components to each pixel necessitate further investigation in future research, along with the exploration of endmember initialization-free methods to enhance model flexibility and reduce reliance on specific initialization assumptions.

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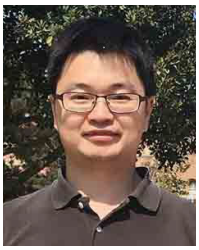
Jianjun Liu (Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence system from Nanjing University of Science and Technology, Nanjing, China, in 2009 and 2014, respectively.

From 2018 to 2020, he was a Post-Doctoral Researcher with the Department of Electrical Engineering, City University of Hong Kong, Hong Kong, China. He is currently an Associate Professor with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China. His research interests are in the areas of hyperspectral image classification, super-resolution, spectral unmixing, sparse representation, computer vision, and pattern recognition.



Hongru Zong received the B.S. degree in environmental science from Nantong University, Nantong, China, in 2022. He is currently pursuing the M.S. degree in computer science and technology with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, China.

His research interests include deep learning and hyperspectral image unmixing.



Zebin Wu (Senior Member, IEEE) received the B.S. and Ph.D. degrees in computer science from Nanjing University of Science and Technology, Nanjing, China, in 2003 and 2008, respectively.

He is currently a Professor with the School of Computer Science, Nanjing University of Science and Technology. His research interests include hyperspectral image processing, high-performance computing, and computer simulation.



Liang Xiao (Senior Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from Nanjing University of Science and Technology, Nanjing, China, in 1999 and 2004, respectively.

From 2009 to 2010, he was a Post-Doctoral Fellow at the Rensselaer Polytechnic Institute, Troy, NY, USA. He is currently a Professor with the School of Computer Science, Nanjing University of Science and Technology. His main research areas include inverse problems in image processing, scientific computing, data mining, and pattern recognition.

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